

UNSUNG WOMEN WARRIORS OF INDIAN SCIENCE

Anna Mani

THE WEATHER WOMAN OF INDIA



K. Venkataraman, Amritanshu Vajpayee,
Nandini Phanse, Shilpa Parikh, Neeru Bhagat

UNSUNG WOMEN WARRIORS OF INDIAN SCIENCE SERIES

ANNA MANI

THE WEATHER WOMAN OF INDIA



(1918-2001)

National Pledge

India is my country.

All Indians are my brothers and sisters.

I love my country,
and I am proud of its rich and varied heritage.

I shall always strive to be worthy of it.

I shall give respect to my parents, teachers
and elders and treat everyone with courtesy.

To my country and my people, I pledge my
devotion.

In their well being and prosperity alone,
lies my happiness.

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THE WEATHER WOMAN OF INDIA: ANNA MANI

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Nandini Phanse, Shilpa Parikh, Neeru Bhagat

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DEDICATED TO ALL THOSE IN THE
PURSUIT OF KNOWLEDGE.

चरन्मार्गान्विजानाति ।

A WANDERER (EVENTUALLY)
FINDS THE PATH.

**उत्तिष्ठत जाग्रत प्राप्य वरान्निबोधत ।
क्षुरस्य धारा निशिता दुरत्यया दुर्गमपथः ॥**

"Arise, awake, and learn by
approaching the noble ones.
The razor's edge is sharp and
hard to cross
— so is the path to wisdom."

CONTENTS

<i>Foreword</i>	<i>ix</i>
<i>Preface</i>	<i>xi</i>
<i>A Note to Young Readers</i>	<i>xv</i>
<i>List of Abbreviations</i>	<i>xvii</i>
<i>List of Figures</i>	<i>xviii</i>
Chapter 01 : The Girl Who Chose Books Over Diamonds	1
Chapter 02 : In the Laboratory of a Nobel Laureate	12
Chapter 03 : From Diamonds to the Sky: A New Path	24
Chapter 04 : Building a Self-Reliant India: The Pune Years - Entering into a New World of Meteorology	31
Chapter 05 : Measuring the Invisible: Ozone and the Sky	41
Chapter 06 : The Sun and the Wind: A Vision for Renewable Energy	49
Chapter 07 : The Scientist and the Human Being	57
Chapter 08 : Honours and Legacy	64
Appendix-A : Timeline of Anna Mani's Life	76
Appendix-B : Major Publications	78
Appendix-C : Glossary of Scientific Terms	80
Appendix-D : Further Reading and Exploration	81

Foreword

In the annals of Indian science, certain names shine with a quiet brilliance, not because they sought the spotlight, but because their work illuminated the lives of millions. Anna Mani was one such soul. Her story is not merely a chronicle of scientific achievement; it is a testament to the power of curiosity, the courage to choose knowledge over convention, and the quiet determination of a woman who refused to let societal norms dictate her destiny.

Born in an era when less than one percent of Indian women were literate, when higher education for girls was considered unnecessary, and when science was viewed as a man's domain, Anna Mani dared to dream differently. At eight years of age, she rejected diamond earrings, the traditional gift for young girls in her community, and asked instead for a set of Encyclopaedia Britannica. That single moment captures the essence of the woman she would become: someone who valued the treasures of the mind far above the ornaments of the body.

This biography is being placed in your hands at a significant moment in India's journey. Our nation, under the vision of Atmanirbhar Bharat, is striving for self-reliance in every sphere, including science and technology. Yet few know that the foundations for self-reliance in Indian meteorology were laid decades ago by a woman in a khadi sari, working tirelessly in a Pune workshop, standardising drawings for nearly a hundred weather instruments. Anna Mani understood that true independence meant more than political freedom; it meant the ability to measure our own skies, predict our own monsoons, and harness our own wind and solar energy.

The authors of this volume have done a remarkable service by bringing Anna Mani's story to young readers. Through these pages, you will encounter not just a scientist, but a human being of extraordinary depth—a woman who slept in her laboratory to record diamond spectra, who led 121 men with quiet authority, who wrote affectionate letters to junior colleagues, and who, even in her seventies, trekked to remote corners of India to set up wind measurement stations. You will meet her students and colleagues, peer into her notebooks, and understand the experiments that changed Indian meteorology. Anna Mani's life serves as a powerful source of inspiration and learning for students.

She teaches us that setbacks are not endings. When denied a Ph.D. for lack of a master's degree, she did not abandon science but found another path to serve her nation. She shows us that excellence requires sacrifice—those twenty-hour days in Raman's laboratory were not in vain. She demonstrates that true leadership is about empowering others—the team she built at the India Meteorological Department continued her work for decades after her retirement.

The National Education Policy 2020 emphasises holistic learning, critical thinking, and the integration of Indian knowledge systems. Anna Mani's life embodies these principles. She was not a narrow specialist but a complete scientist—equally at home with diamond spectroscopy and wind energy surveys, with ozone measurements and solar radiation networks. Her work bridged the gap between fundamental physics and practical application, between the laboratory and the service of the nation.

As you read this book, we invite you to reflect on Anna Mani's own words: "We have only one life. First equip yourself for the job, make full use of your talents and then love and enjoy the work, making the most of being out doors and in contact with nature." These are not just words of advice; they are the philosophy by which she lived. She never married, not because she was opposed to marriage, but because she believed one life was too short to pursue all the knowledge she sought.

In an age of instant gratification and fleeting attention, Anna Mani's story reminds us of the value of sustained effort, of patient observation, of work done not for recognition but for its own sake. She received many honours, the INSA K.R. Ramanathan Medal, and fellowships of several prestigious academies, but never sought them. When Google honoured her with a doodle on her 104th birth anniversary, it was not because she had chased fame, but because her work had quietly shaped the lives of billions.

To the young readers who will journey through these pages: let Anna Mani's story inspire you to ask your own questions, to seek your own answers, and to never let anyone tell you that your dreams are too big or your aspirations misplaced. Whether you are a girl wondering if science is for you, or a boy wondering if dedication matters, Anna Mani's life provides an answer: yes, it does. Yes, you can. Yes, the pursuit of knowledge is always worthwhile.

May this biography light a spark in your heart, just as Anna Mani's own spark was lit by those encyclopaedias so many years ago.

[Eminent Personality]

Preface

Every great scientist was once a child with a question. For Anna Mani, that question was not simply about how the world worked, but about how she could use her understanding to serve her nation. This biography is the story of that child—her growth, her struggles, her triumphs, and her transformation into one of India's most remarkable scientists.

Anna Mani's journey begins in the lush hills of Travancore, in a large Syrian Christian family where tradition held sway. But from her earliest years, Anna was different. While her sisters prepared for the roles society had assigned them, Anna buried herself in books, reading everything she could find in Malayalam and English. By the time she was twelve, she had exhausted the resources of her local library, a feat that would have been remarkable anywhere, but was considered even more extraordinary in the context of early twentieth-century India.

Her decision to ask for encyclopaedias instead of diamond earrings on her eighth birthday has become legendary, and rightly so. It was not merely the act of a precocious child, but the first public declaration of a philosophy she would carry throughout her life: that knowledge was the only ornament worth having. This philosophy would sustain her through the challenges that lay ahead—through the male chauvinism in the educational institutions and research laboratory, through the denial of her Ph.D., through the long years of building India's meteorological infrastructure from almost nothing.

The Anna Mani you will meet in these pages is not a distant, marble statue of a scientist, but a living, breathing human being. She was the woman who slept on a bench in the laboratory because her diamond samples needed twenty hours of exposure. She was the leader who refused to transfer junior colleagues to Delhi during winter because they lacked warm clothing. She was the mentor who defended a writer against colleagues who mocked his literary pursuits. She was the friend who wrote long, affectionate letters describing not just scientific work but the antics of her maid and the beauty of foreign cities.

These human details matter because they remind us that science is not separate from life, but an integral part of it. Anna Mani was not only a great scientist but was also a passionate human. Her love of nature, her joy in trekking and bird-watching, her delight in music and books—all these fed into her scientific work, giving her the patience to observe, the creativity to innovate, and the wisdom to see connections that others missed.

Central to Anna Mani's legacy is her contribution to making India self-reliant in meteorological instrumentation. When she joined the India Meteorological Department in 1948, the country depended entirely on imported instruments, mostly from Britain. Simple devices like thermometers and barometers were bought from abroad; complex instruments like radiosondes were beyond Indian capability. Anna Mani changed this. Working with a visionary leader, S.P. Venkiteshwaran, she built a world-class instrument manufacturing facility from scratch. She standardised drawings for nearly a hundred instruments, trained a team of scientists and technicians, and ensured that every device met rigorous quality standards.

But her vision extended far beyond instrumentation. She recognised early the importance of solar energy for a tropical country like India, and established a network of radiation monitoring stations that provided data still used today. She began studying atmospheric ozone in 1960, years before the world understood the ozone layer's critical role in protecting life on Earth. She developed the Indian ozonesonde, which detected the ozone hole over Antarctica three years before the Nobel Prize-winning work of Rowland and Molina. She pioneered wind energy assessment in India, organising measurements at hundreds of sites and publishing volumes that became a guidebook for wind farm developers.

As you read this book, you will encounter not just these achievements, but the context in which they were accomplished. Colonial India offered few opportunities for Indian scientists, and even fewer for women. The literacy rate for women when Anna was born was less than one percent. The number of women enrolled in colleges was under a thousand. Against this backdrop, Anna Mani's journey from a book-loving girl in Travancore to Deputy Director General of the India Meteorological Department is nothing short of miraculous.

The authors of this volume have drawn upon a wealth of sources to bring Anna Mani's story to life: her scientific papers, her personal letters, interviews with colleagues, archival photographs, and the recent biography by Asha Gopinathan. We have also been guided by the example of the book on Satyendra Nath Bose, which showed how a scientific biography can be both academically rigorous and deeply humane. Our goal has been to create a book that students will read not as a textbook, but as a story—a story of curiosity, courage, and commitment.

For young readers, Anna Mani's life offers a treasure trove of lessons. She teaches us that setbacks are opportunities in disguise—when denied a Ph.D.,

she found her true calling in meteorology. She shows us that leadership is about service—she led 121 men not by shouting, but by example. She demonstrates that science is not just about equations but about observation—her love of nature made her a better meteorologist. She reminds us that simplicity and integrity matter—she wore khadi all her life and never sought the limelight.

In an age of climate change, when understanding the atmosphere has never been more critical, Anna Mani's work takes on new significance. The solar radiation data she collected, the wind energy maps she created, the ozone measurements she pioneered—all these remain essential tools for scientists today. Her vision of self-reliance in science, of using indigenous knowledge and local manufacturing, resonates powerfully with current national priorities.

But ultimately, this biography is not about policies or priorities. It is about a person—a remarkable, determined, brilliant person who happened to be a woman, who happened to be Indian, who happened to love the sky. It is about the choices she made, the obstacles she overcame, and the legacy she left. It is about why Anna Mani matters.

We invite you to join us on this journey through Anna Mani's life. May it inspire you as much as it has inspired us.

K. Venkataraman
Amritanshu Vajpayee
Nandini Phanse
Shilpa Parikh
Neeru Bhagat

A Note to Young Readers

“You can, but do your best.”

Anna Mani

Dear Student,

This book is about a remarkable woman named Anna Mani. But more than that, it is about questions.

When Anna was your age, she asked questions that others thought unnecessary. Why should girls only receive jewellery? Why should boys be educated and not girls? Why should India depend on other countries for simple instruments? Why should we not measure our own weather?

These questions led her on an extraordinary journey—from a hillside in Kerala to the laboratories of Bangalore, from the workshops of Pune to the scientific conferences of Europe and America. Along the way, she discovered not just answers, but a purpose: to make India self-reliant in understanding and exploring its own skies.

As you read this book, you will encounter some challenging concepts—spectroscopy, radiation measurement, ozone chemistry. Do not be discouraged if you do not understand everything immediately. Science is not about knowing all the answers, but about asking good questions. Anna herself spent years learning, experimenting, failing, and trying again.

We have included special features to help you on this journey:

- Try to Imagine boxes invite you to put yourself in Anna's place
- Think Like a Scientist sections explain key concepts simply
- Mini-Profiles introduce you to people who influenced Anna or were influenced by her
- Breakout Boxes explore interesting side topics

Use these features. Discuss them with your friends and teachers. Let Anna's questions become your questions. Let her curiosity spark your own. Remember: Anna Mani was not born a scientist. She became one through

years of hard work, through setbacks that would have defeated a lesser spirit, through a determination to serve her country. You too can follow this path—not necessarily to become a scientist, but to become someone who asks questions, who seeks answers, who refuses to accept that things must remain as they are.

So turn the page. Begin the journey. And let Anna Mani's story inspire your own.

List of Abbreviations

B. Sc.	– Bachelor of Science
CFCs	– Chlorofluorocarbons
DST	– Department of Science and Technology
IGY	– International Geophysical Year
IISc	– Indian Institute of Science
IMD	– India Meteorological Department
INSA	– Indian National Science Academy
ISBN	– International Standard Book Number
NPL	– National Physical Laboratory
Ph.D.	– Doctor of Philosophy
RRI	– Raman Research Institute
SI	– International System of Units
UK	– United Kingdom
UN	– United Nations
WMO	– World Meteorological Organization

List of Figures

1. Figure a: Statue of Anna Mani, Birla Industrial & Technological Museum, Kolkata, West Bengal, India.
2. Figure 1.1: Anna (left) with a friend, Mariamma Kochamma.
3. Figure 2.1: A young researcher – Anna Mani.
4. Figure 2.2: Sir C.V. Raman and his wife, Lokasundari Ammal.
5. Figure 2.3: A page from 'Anna's Extraordinary Experiments with Weather'.
6. Figure 3.1: Anna Mani with friends in the U.K. around 1945.
7. Figure 4.1: India Meteorological Department, Pune.
8. Figure 4.2: Anna Mani at WMO CIMO-V Versailles 1969.
9. Figure 4.3: Raingauge @IMD Pune.
10. Figure 4.4: Anna Mani in Payeme, Switzerland, 1956.
11. Figure 4.5: Anna Mani at the International Radiation Symposium, Sendai, Japan, 1972.
12. Figure 5.1: Anna Mani and a colleague work on a radiosonde, a balloon-borne weather-measuring equipment.
13. Figure 5.2: Anna Mani with international scientists in Tübingen, Germany, in 1965.
14. Figure 6.1: Anna Mani at an event.
15. Figure 7.1: Anna Mani Farewell, IMD Poona, 1969.
16. Figure 7.2: Anna Mani with her dogs Jenny and Sherry in Delhi.
17. Figure 8.1: Anna Mani (Left) with Satish Dhawan (Second from Left), who guided the development of India's space programme in its formative years.
18. Figure 8.2: Participants to the 47th Annual Meeting of the Indian Academy of Sciences at Trivandrum, 5 – 8 November 1981.
19. Figure 8.3: Anna Mani at the Indian Academy of Sciences Golden Jubilee Meet in Bangalore, 1984.
20. Figure 8.4: Google Doodle: Anna Mani's 104th Birthday.
21. Figure 8.5: Anna Mani (sitting right to Sir C.V. Raman) with stalwarts of science and Technology in modern India during the 80th Birthday celebration of Sir C.V. Raman, 1968.



(Figure a: Statue of Anna Mani, Birla Industrial & Technological Museum, Kolkata, West Bengal, India)

CHAPTER 1

The Girl Who Chose Books Over Diamonds

"Don't bother me with jewellery; let me read."
- Anna Mani

A Hillside Childhood in Travancore

The Western Ghats of India are among the most beautiful places on Earth. Lush green hills roll like waves toward the sky, covered with evergreen tropical rainforests that have stood for millennia. Perennial rivers tumble down rocky slopes, and the air is filled with the calls of exotic birds and the rustle of wildlife. On the western side, the Arabian Sea laps at a long coastline, completing a landscape of extraordinary diversity.

It was here, in the princely state of Travancore (now part of Kerala), that Anna Modayil Mani was born on August 23, 1918. The place was Peermade, also spelt Peerumedu (in Kerala's present Idukki district), a hill station known for its tea and cardamom plantations, its cool climate, and its breathtaking views. Anna would later describe Travancore as "one of the most beautiful places on earth", a sentiment born of childhood memories that never faded.

The Mani family belonged to the ancient Syrian Christian community of Kerala, a community that traced its origins to the arrival of St. Thomas the Apostle in AD 52. This heritage carried with it a long tradition of education and learning. Yet even within this relatively progressive community, the roles of men and women were clearly defined. Men were educated for careers; women were prepared for marriage and motherhood.

Anna's father, Modayil Pothen Mani, was a civil engineer in the Travancore Public Works Department, responsible for building roads and bridges. He was, by Anna's own account, an agnostic, unusual for his time and community, and his rational, questioning attitude deeply influenced his

A Glimpse into Anna's Family	
Family Member	Details
Father	Modayil Pothen Mani, civil engineer, agnostic, rationalist
Mother	Ammamma
Siblings	Five brothers, two sisters (Anna was seventh of eight)
Community	Syrian Christian, ancient tradition, educated
Home Location	Peermade, Travancore (now Kerala), Western Ghats

youngest daughter. He taught her to accept nothing on faith, to test every statement, to prove everything for herself. This scientific temperament, instilled in childhood, shaped Anna’s entire life.

Her mother, Ammama, was a source of inspiration for her, though details of her life are scarce in the historical record. What we know is that she managed a household of ten people—Anna’s parents, their eight children, and the extended family that often lived with them. Anna was the seventh of those eight children: five boys and three girls.

Holidays in Anna’s childhood were spent mostly in the mountains, where her father owned cardamom estates. Young Anna would wander through the forests, studying trees, watching birds, observing wild animals with the patient attention that would later characterize her scientific work. This early exposure to nature was not merely recreational- it was the beginning of her education in observation, in noticing details, in understanding patterns.

The cardamom estates were places of wonder. The spice itself—known as the “queen of spices”—grew in the shade of taller trees, its pods hidden among broad leaves. Anna learned to recognise the plants, to understand the cycle of growth and harvest, and to appreciate the complexity of the ecosystem. She watched birds build nests and raise young, observed monkeys swing through the canopy, and tracked the movements of smaller creatures through the undergrowth.

These childhood experiences gave Anna something that no classroom could provide: a direct, personal connection to the natural world. She understood, in a way that only those who have spent time in nature can, that the world is not a collection of separate objects but a web of relationships.

Try to Imagine

Close your eyes and imagine yourself in the Western Ghats of Kerala, nearly a hundred years ago. You are eight years old. Around you are hills covered with tea plantations and rainforest. The air is cool and moist, smelling of earth and spice. Birds call from every direction—you cannot see them, but you hear their songs. Monkeys crash through the branches above.

You are Anna Mani, and this is your playground. There are no smartphones, no televisions, no computers. Your entertainment is the natural world itself. Every day, you discover something new—a flower you haven't seen before, a bird you cannot identify, a pattern in the clouds that you don't understand.

How does this childhood shape you? What questions do you ask? What mysteries do you want to solve?

Every plant, every animal, every drop of water is connected to everything else. This holistic understanding would later inform her work in meteorology, where she studied the atmosphere as an integrated system rather than a collection of isolated phenomena.

The Eighth Birthday That Changed Everything

In the Mani household, as in most traditional Indian families, birthdays were occasions for celebration. But there was a particular custom that Anna had come to dread. On their eighth birthdays, the girls of the family received a special gift: a pair of diamond earrings.

This was not mere ornamentation. In the community's tradition, diamonds were more than jewellery—they were a form of security, a portable wealth that a woman could take with her when she married. The earrings symbolised a girl's transition toward womanhood and her eventual role as a wife and mother. They were, in a very real sense, an investment in her future.

Anna wanted none of it.

As her eighth birthday approached in 1926, Anna found herself dreading the inevitable ceremony. She had watched her older sisters receive their earrings, had seen them displayed and admired, had heard the adults discuss their quality and value. None of it interested her. What interested her were books.

By this age, Anna had already read every Malayalam book she could find in her home and in the local library. She had moved on to English books, struggling through texts meant for much older readers. Her hunger for knowledge was insatiable, and the world of books had opened doors she never knew existed.

“Don’t bother me with jewellery; let me read,” she told her family when they mentioned the approaching birthday.

Her parents were initially taken aback. The gift of earrings was not merely a tradition; it was a social expectation. What would relatives think if the girl received no jewellery? What would they say about the family?

But Anna was firm. She did not want gold, silver, or diamonds. She wanted something else entirely: a set of the Encyclopaedia Britannica.

Think Like a Scientist: What is an Encyclopaedia?

Today, if you want to know something, you type a question into a search engine and get millions of answers in seconds. But in 1926, information was not so easily accessible. Encyclopaedias were among the few comprehensive sources of knowledge available to curious minds.

The Encyclopaedia Britannica was the most famous of these. First published in Edinburgh, Scotland, in 1768, it had grown over a century and a half into a monumental work—dozens of volumes containing articles on every subject imaginable, written by the world’s leading experts. Owning a set was like having a university in your home.

For a girl like Anna, who had exhausted the resources of her local library, the Encyclopaedia Britannica represented an entire universe of knowledge waiting to be explored. It was a window into worlds she could never visit, into sciences she had barely heard of, into history, geography, literature, and art. It was the most precious gift she could imagine.

Her father, the rationalist engineer, understood. Here was his daughter, not asking for adornment but for knowledge. Not following tradition but forging her own path. Not accepting what was given but demanding what she truly wanted.

He arranged for the encyclopaedia to be delivered.

On her birthday, Anna tore open the gift and peered inside. No diamonds sparkled there. Instead, volume after volume of the Encyclopaedia

Britannica sat in neat rows, their covers promising adventure, discovery, and learning. Anna was overjoyed.

“So many books. Hurray!” she shouted, running around the house to hug everyone.

This moment, the eighth birthday when Anna chose books over diamonds, became legendary in her family and, later, in the story of Indian science. It was the first public declaration of a philosophy she would carry throughout her life: that knowledge was the only ornament worth having, that learning was the greatest adventure, that the treasures of the mind far exceeded the treasures of the body.

A Library of One’s Own

The encyclopaedia was just the beginning. Anna’s hunger for books only grew with feeding.

In Peermade, the public library became her second home. She would spend hours there, moving from shelf to shelf, pulling down books that caught her attention, reading until the librarian announced closing time. The librarians, initially amused by this earnest little girl, soon came to respect her determination. They would set aside books they thought she might like, recommend authors she hadn’t discovered, and ensure she had access to the best materials available.

By the age of twelve, Anna had read every Malayalam book in the library—every novel, every collection of poetry, every work of non-fiction.

Books That Shaped Young Anna	
Type of Book	What young Anna Mani gained
History	Understanding of how societies change, role of individuals
Geography	Appreciation of diversity, connection to wider world
Science	Explanation of natural phenomena, methods of inquiry
Biography	Models of achievement, examples of overcoming obstacles
Literature	Imagination, empathy, understanding of human nature

She had also made substantial inroads into the English section, tackling authors whose works were intended for adult readers.

What did she read? Everything. History and geography opened windows to distant lands and past times. Biographies of great men and women showed her what was possible. Works of science explained the natural world she loved so much. Novels and poetry fed her imagination and taught her about human nature.

But perhaps most importantly, the books gave Anna something she could find nowhere else: a sense of possibility. In a society where girls were expected to marry young and raise families, the books showed her women who had done extraordinary things—scientists, explorers, writers, reformers. They showed her that the path society had laid out was not the only path. They gave her the courage to imagine a different life.

The books also gave her something more subtle: a deep sense of social justice. Reading about the struggles of ordinary people, about exploitation and inequality, about movements for reform and revolution, Anna developed a conviction that the world could and should be better. This conviction would later inform her work in independent India, where she dedicated herself to building scientific institutions that would serve all citizens, not just the privileged few.

Her family, though initially puzzled by this bookish child, came to support her passion. Her father, already inclined toward rationalism, encouraged her reading. Her mother, despite her traditional background, ensured that Anna had time and space for her books. Even the servants in the household learned to leave Anna alone when she was reading—she was not to be disturbed.

Mini-Profile: The Librarian of Peermade

History has not recorded the name of the librarian who presided over the Peermade public library in the 1920s. But his or her role in Anna Mani's development cannot be overlooked.

In that small library in a hill station of Travancore, a young girl found not just books but encouragement. The librarian noticed this earnest child who came day after day, who read hungrily, who asked for recommendations. Instead of dismissing her as a nuisance, the librarian nurtured her curiosity—setting aside books, answering questions, opening doors to new worlds.

Librarians are often unsung heroes of intellectual development. They are the gatekeepers of knowledge, the guides through the wilderness of

information. The Peermade librarian, whoever he or she was, recognised a spark in young Anna and fanned it into flame.

When Anna later became a scientist, when she published her first papers, when she represented India on the world stage, she carried with her the gift that the librarian had given: the belief that knowledge was hers for the taking, that the library was a palace where she was always welcome, that the world of books was a world of infinite possibility.

Education

Anna Mani's father's profession required the family to relocate frequently, leading her to change schools often. She began her education at His Highness the Maharaja's School for Women in Tiruvananthapuram, and later studied at Christava Mahilalayam School for Girls in Alwaye. For higher studies, she moved to Madras (now Chennai), where she spent her first two years at Pachaiyappas College, followed by a three-year honours program in Chemistry and Physics at Presidency College.

Think Like a Physicist: Why Physics?

Physics is the most fundamental of the sciences. It asks the biggest questions: What is the universe made of? How do things move? What are the forces that shape reality? From the smallest particles to the largest galaxies, physics seeks to understand the rules that govern everything.

For Anna, physics offered a way to understand the natural world she loved. Why is the sky blue? (Rayleigh scattering.) How do clouds form? (Thermodynamics and fluid dynamics.) What causes wind? (Pressure differences and the Coriolis effect.) Physics provided answers to these questions, and also raised new ones.

But physics also offered something else: a challenge. It was difficult, demanding, and required rigorous thinking. Anna was not afraid of difficulty; she welcomed it. The harder the problem, the more satisfying the solution. "There was a consensus at that time that education for women should be tailored to their particular roles as mothers and homemakers," Anna later recalled. But she refused to accept that consensus. Physics was not considered a suitable subject for women, all the more reason to pursue it.

Choosing Physics

As Anna completed her schooling, the question of her future became crucial. What would she do with her life?

In her community, the path for girls was clear: they married, usually in their late teens, and devoted themselves to home and family. Anna's older sisters had followed this path. Their parents expected nothing different for Anna.

But Anna had other ideas. She wanted to study further. She wanted to pursue higher education. She wanted, though she could not yet articulate it clearly, to become a scientist.

Her family was not opposed to education—her father's rationalism and her own evident brilliance argued in her favour. But what kind of education? What subject?

Initially, Anna considered medicine. The idea of healing people, of using science to relieve suffering, appealed to her. But as she thought more about her interests and aptitudes, she realised that physics drew her more strongly. The laws that governed the universe, the forces that shaped matter and energy, the elegant mathematics that described it all, these were the questions that excited her.

In 1935, Anna enrolled at Pachaiyappas College in Madras (now Chennai) for her intermediate studies. Two years later, she moved to Presidency College, one of the most prestigious institutions in India, to pursue a Bachelor of Science degree with honours in physics and chemistry.

Presidency College was a revelation. Here, for the first time, Anna encountered students and teachers who shared her passion for science. The laboratories were well-equipped, the library excellent, and the intellectual atmosphere electric. Professor H. Parameswaran taught physics with clarity and enthusiasm; Professor B.B. Dey did the same for chemistry. Anna threw herself into her studies, spending long hours in the laboratory, mastering experimental techniques, and developing the hands-on skills that would later prove invaluable.

In 1939, Anna graduated with her B.Sc. (Honours) degree. She had done well, though not spectacularly, her grades were good but not exceptional. What mattered more was what she had learned: how to think scientifically, how to design experiments, how to interpret data, how to persist in the face of difficulty.

The Khadi Clad Student

The 1920s and 1930s were decades of extraordinary ferment in India. The freedom struggle, led by Mahatma Gandhi, was gaining momentum. Across the country, Indians were questioning British rule and demanding self-governance. The call for *Swaraj*, self-rule, echoed in towns and villages, in schools and colleges, in the hearts of millions.

Anna Mani was too young to participate directly in political activities, but the nationalist movement reached her nonetheless. In 1924, when she was six years old, Gandhi visited Travancore and led the Vaikom Satyagraha, a movement demanding the right of lower-caste Hindus to use roads near the Vaikom temple. The satyagraha captured the imagination of the region, and Anna, even at that young age, was deeply impressed.

For a six-year-old girl like Anna, witnessing Gandhi's visit and the passion of the satyagrahis was a formative experience. She saw that ordinary people could challenge unjust systems. She saw that non-violent resistance could achieve what violence could not. And she saw a vision of India that was free not just from British rule, but from its own internal divisions.

One of Gandhi's most powerful symbols was *khadi*, the hand-spun, hand-woven cloth. By wearing khadi, Indians could reject British-manufactured textiles and support local artisans. Khadi became a symbol of self-reliance, of dignity, of resistance to colonial exploitation.

Anna was captivated. She began wearing only khadi garments, a practice she continued throughout her life. The coarse, hand-spun cloth was not as

Think Like a Historian: The Vaikom Satyagraha

The Vaikom Satyagraha (1924-1925) was a landmark event in India's social history. It was the first time that the demand for temple entry for all Hindus, regardless of caste, was raised through organised non-violent protest. The movement took place in Vaikom, a town in Travancore (now in Kerala), and attracted national attention.

Gandhi himself visited Vaikom during the satyagraha, supporting the protesters and negotiating with authorities. The movement eventually achieved a partial victory—certain roads around the temple were opened to all castes. More importantly, it set a precedent for future temple entry movements across India.



*(Figure 1.1: Anna (left) with a friend, Mariamma Kochamma. Image Courtesy: Anna Phil,
Source: <https://annamani.co.in/education.html>)*

comfortable or elegant as the fine textiles her family could afford. But for Anna, it represented something more important: commitment to her country, solidarity with the poor, and participation in a movement larger than herself.

The khadi was also a statement about simplicity. Anna was learning that a meaningful life did not require luxury or display. What mattered was purpose, dedication, service.

These values would guide her through decades of scientific work, through the challenges of building institutions, through the recognition that eventually came her way.

For Anna Mani, wearing khadi was not just a political statement. It was a commitment to simplicity, to living with less, to focusing on what truly mattered. Throughout her life, she wore khadi saris, rejecting the fine fabrics and jewellery that her family's wealth could have afforded. Her ornament was her work; her luxury was her science.

Breakout Box: What is Khadi?

Khadi is cloth that is hand-spun on a charkha (spinning wheel) and hand-woven. The word comes from khaddar, which means hand-spun cloth in several Indian languages.

Mahatma Gandhi made khadi a symbol of the Indian independence movement. He argued that by spinning their own cloth, Indians could:

- Reject British manufactured textiles
- Support rural artisans and weavers
- Demonstrate self-reliance (swadeshi)
- Practice simplicity and manual labour

Gandhi himself spun regularly and wore only a simple khadi loin cloth. He encouraged all Indians to do the same, making the spinning wheel a central symbol of the freedom struggle.

CHAPTER 2

In the Laboratory of a Nobel Laureate

"He (Prof. C.V. Raman) achieved in one lifetime what others would take many lives to accomplish."

- Anna Mani

The Presidency College Days

Before we follow Anna to Bangalore and the laboratory of C.V. Raman, we must pause to understand what Presidency College meant to a young woman like her in the 1930s.

Presidency College was not just any institution. Founded in 1840, it was one of the oldest and most prestigious colleges in India. Its alumni included some of the country's greatest minds, scientists, writers, politicians, reformers. The college had been at the forefront of higher education in South India for a century, and its standards were correspondingly high.

For Anna, arriving at Presidency College in 1937 was like entering a new world. The campus, with its red brick buildings and sprawling grounds, radiated history and scholarship. The library, with its thousands of volumes, promised endless exploration. The laboratories, equipped with instruments imported from Europe, offered hands-on experience with real science.

But perhaps most important were the teachers. Professor H. Parameswaran, who taught physics, was known for his clarity and his ability to make complex concepts accessible. Professor B.B. Dey, who taught chemistry, was equally knowledgeable. Both recognized Anna's potential and encouraged her to push beyond the curriculum, to ask questions, to think independently.

Prof. Parameswaran was more than a teacher. He was a mentor who took a personal interest in his students' development. He noticed Anna's dedication, her careful experimental technique, and her willingness to spend extra hours in the laboratory. He encouraged her to apply for research



*(Figure 2.1: A young researcher – Anna Mani.
Photo credit: Raman Research Institute Digital repository.)*

positions, wrote letters of recommendation, and followed her career with interest long after she left the college.

Years later, when Anna had become a renowned scientist, she would credit Parameswaran with giving her the confidence to pursue research. He had shown her that she belonged in science, that her gender was irrelevant to her ability, that the only thing that mattered was the quality of her work.

The laboratory work at Presidency College was particularly valuable. Anna spent hours mastering experimental techniques- learning to handle delicate instruments, to make precise measurements, to record observations systematically. She discovered that she had a gift for working with her hands,

Mini-Profile: Professor H. Parameswaran

Professor H. Parameswaran was one of the most influential physics teachers in South India during the 1930s and 1940s. A graduate of Presidency College himself, he had studied at Cambridge and returned to India to teach. His lectures were legendary for their clarity, students said he could explain quantum mechanics to a very young guy as well.

for coaxing recalcitrant equipment to perform, for noticing small details that others missed.

This hands-on aptitude would prove crucial in her later career. While many physicists focus entirely on theory, Anna was equally comfortable with practical work. She could design an instrument, build it, calibrate it, use it, and interpret the results. This combination of theoretical understanding and practical skill made her uniquely valuable.

Experimental vs. Theoretical Physics

Physics is often divided into two broad approaches: theoretical and experimental.

Theoretical physicists work with mathematics and concepts. They develop equations that describe how the universe works, make predictions about phenomena that haven't been observed, and interpret experimental results in light of fundamental principles. Think of them as architects, designing the grand structure of understanding.

Experimental physicists work with instruments and measurements. They design experiments to test theories, develop new techniques for observing phenomena, and gather data that can confirm or contradict theoretical predictions. Think of them as builders, constructing the evidence that supports the theoretical architecture.

Both approaches are essential, and the greatest physicists are often skilled at both. Anna Mani was one of those—a scientist who could move seamlessly between theory and experiment, who understood both the mathematical foundations and the practical requirements of her work.

At Presidency College, Anna was laying the foundation for this dual capability. She was learning the theory from professors like Parameswaran, and she was developing experimental skills in the laboratory. Both would serve her well in the years to come.

A Scholarship to Bangalore

In 1940, Anna received news that would change her life: she had been awarded a research scholarship to study at the Indian Institute of Science (IISc) in Bangalore.

The Indian Institute of Science was, and remains, one of India's premier research institutions. Founded in 1909 with support from Jamsetji Tata and the Mysore royal family, it was designed to be India's answer to the great research universities of the West. Its faculty included some of the country's

most distinguished scientists, and its laboratories were among the best-equipped in Asia.

At the heart of the physics department was a man who had already achieved legendary status – none other than our Sir C.V. Raman. C.V. Raman had won the Nobel Prize in Physics in 1930 for his discovery of the *Raman Effect*, a phenomenon in which light changes wavelength when scattered by molecules. This discovery had put Indian science on the global map and made Raman a national hero. He was also director of the IISc Physics department, and he ran his laboratory with an iron hand.

Mini-Profile: Kamala Sohonie

Kamala Sohonie (1912-1998) was a pioneer who paved the way for women like Anna Mani. When she applied to IISc in 1933, she was refused an admission in the beginning. It was only after her father wrote a letter and Lokasundari Ammal intervened, she was admitted to the institution conditionally on probation.



(Figure 2.2: Sir C.V. Raman and his wife Lokasundari Ammal, Image Source: Internet)

Sohonie proved herself spectacularly. Within a year, her work on plant proteins was so impressive that C.V. Raman extended her scholarship and relaxed some restrictions. She went on to earn her Ph.D. from Cambridge University and made significant contributions to biochemistry, including research on the nutritional value of different varieties of rice. Sohonie later became director of the Institute of Science in Bombay and received numerous honours for her work.

For a young physicist like Anna, the chance to work under Raman was extraordinary. It was like a young painter being invited to study with Michelangelo, or a young musician being chosen to learn from Beethoven. Raman was not just a scientist; he was an institution, a force of nature, a man whose very presence demanded excellence.

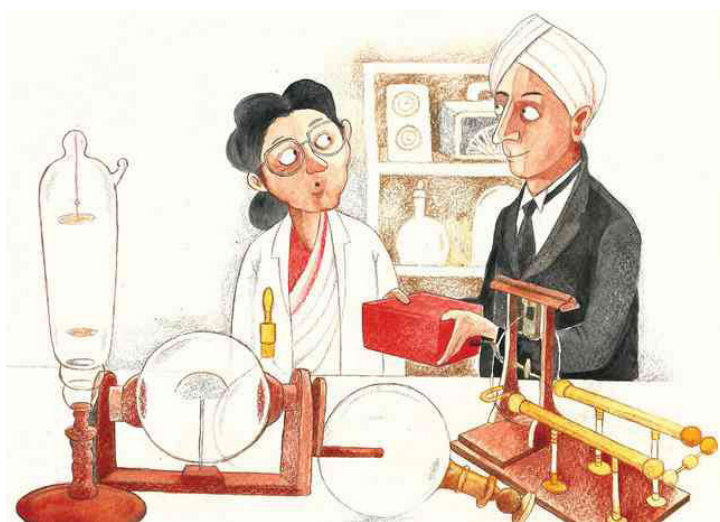
Hence, the opportunity for young Anna Mani came with challenges. Raman was academically demanding, expecting his students to work long hours and accept nothing less than perfection. In those days, in general, women in the society were expected to prove themselves constantly, while men were given the benefit of the doubt.

After the academic success of Kamala Sohonie, a brilliant biochemist who would later become the first Indian woman to earn a Ph.D. in a scientific field, IISc began to open its doors more seriously to women scientists. By 1940, when Anna arrived, Raman had accepted that women would be part of his laboratory. But the atmosphere remained difficult.

Anna Mani, arriving at IISc seven years after Sohonie, benefited from the doors that Sohonie had forced open. But the atmosphere remained challenging, and Anna had to prove herself daily.

Raman's Domain: The World of Light and Crystals

What was it like to work in Raman's laboratory?



*(Figure 2.3: A page from 'Anna's Extraordinary Experiments with Weather'.
Image credit: Nandita Jayaraj & Priya Kurian, Pratham Books.)*

Imagine a large room filled with optical benches, spectrographs, and other instruments for studying light. Sunlight streams through windows, but much of the work is done in darkness, because light measurements require precise control. Students huddle over their apparatus, making minute adjustments, recording observations, and developing photographic plates.

Raman himself moves through the laboratory like a general inspecting troops. He stops at each student's setup, asks pointed questions, and offers criticism or praise in equal measure. He is everywhere at once, noticing everything, missing nothing. His presence fills the room.

The focus of the laboratory was spectroscopy—the study of how light interacts with matter. When light passes through a material, some wavelengths are absorbed, some are transmitted, and some are scattered. The pattern of absorption and scattering reveals information about the material's structure, its chemical composition, and its physical properties.

Raman had won his Nobel Prize by discovering that scattered light could change wavelength, a phenomenon now called Raman scattering. His laboratory continued to explore this and related effects, focusing particularly on crystals: diamonds, rubies, and other precious stones.

Think Like a Scientist: What is Spectroscopy?

Spectroscopy is one of the most powerful tools in physics and chemistry. Here's how it works:

1. You shine light on a sample.
2. Some of that light is absorbed by the sample's atoms or molecules.
3. The absorbed energy causes electrons to jump to higher energy levels.
4. When the electrons fall back down, they emit light at specific wavelengths.
5. The pattern of absorbed and emitted wavelengths, the spectrum, acts like a fingerprint, revealing what the sample is made of and how its atoms are arranged.

Different materials produce different spectra. A diamond's spectrum is different from a ruby's. A pure diamond's spectrum is different from one with impurities. By studying these spectra carefully, scientists can learn about the structure and properties of materials without destroying them.

For Raman's group, spectroscopy was a way to explore the fundamental nature of crystals. Diamonds were particularly interesting because they are pure carbon arranged in a perfect lattice. Any deviation from perfection—any impurity, any defect—showed up in the spectrum. By studying diamonds with different colours, different origins, and different histories, Raman's students could learn about the conditions under which they formed.

Sleeping in the Laboratory

Anna's first project was to study the fluorescence and absorption spectra of rubies. Rubies are Aluminium oxide with a small amount of chromium—the chromium atoms give the stone its red colour and also produce characteristic patterns in its spectrum.

The work required immense patience. Ruby samples often had weak luminescence, meaning they emitted very little light. To record their spectra, Anna had to expose photographic plates for hours, sometimes fifteen to twenty hours at a stretch.

This created a practical problem. The laboratory closed at night, but the exposures couldn't be interrupted. Once a plate was exposed, it had to remain in place until the exposure was complete. Opening the apparatus would ruin the measurement. Anna's solution was simple: she stayed with her experiments. Night after night, she slept in the laboratory. A bench served as

Try to Imagine

You are twenty-two years old, alone in a dark laboratory in Bangalore. Outside, the city sleeps. Inside, your apparatus hums quietly, exposing a photographic plate that will record the secret language of a ruby. You have fifteen hours to wait.

What do you think about? Your family in Kerala? Your experiments? The future that awaits you? Do you feel lonely, or does the solitude give you space to think? Are you anxious about whether the exposure will work, or confident in your technique?

Anna Mani spent many nights like this. She emerged from them with data that would become five published papers. She also emerged with something else: a deep understanding of her materials, a patience for long experiments, and a determination that no obstacle was too great.

her bed, a lab coat as her blanket. The long hours of darkness gave her time to think, to plan the next day's work, to reflect on what the spectra might mean. She learned to read the faint traces on photographic plates, to distinguish signal from noise, to recognise patterns that others might miss.

The work on rubies was successful. Anna produced clear spectra showing the fluorescence patterns of different ruby samples. She learned to relate these patterns to the crystal structure, to identify the signatures of chromium atoms, and to understand how temperature and other conditions affected the spectra.

Then Raman assigned her a new project – diamonds, yes, the diamonds indeed!

Diamonds were even more challenging than rubies. Their luminescence could be extremely weak, requiring even longer exposures. But they were also more scientifically interesting. In the early 1940s, there was an ongoing debate about the structure of diamonds and the nature of their optical properties.

Raman himself had strong views on this debate. He believed that diamonds had a quasi-periodic structure—not perfectly regular, but with subtle variations that affected their properties. This put him in opposition to Max Born, one of the giants of theoretical physics, who argued for a perfectly regular lattice.

Anna's work on diamonds would contribute evidence to this debate.

The Raman-Born Controversy

In the 1940s, two of the world's greatest physicists disagreed about the structure of diamonds.

C.V. Raman argued that diamonds had a quasi-periodic structure—that the carbon atoms were not arranged in a perfectly regular lattice, but with subtle variations that repeated in complex patterns. This, he believed, explained certain features of diamond spectra.

Max Born (another Nobel laureate) argued for a perfectly regular lattice. His mathematical models suggested that diamond's properties could be explained without assuming any irregularity.

The controversy was intense and sometimes personal. Each scientist defended his position vigorously, and their students and collaborators took sides.

Anna Mani's experimental work on diamond spectra provided data relevant to this debate. Her measurements of fluorescence and absorption patterns helped illuminate the actual structure of diamonds, contributing evidence that both sides had to consider.

Science advances through such debates. Disagreement forces researchers to refine their arguments, to collect better data, to think more deeply. The Raman-Born controversy, though never fully resolved, pushed forward our understanding of crystal physics.

The Diamonds That Spoke

Between 1942 and 1945, Anna Mani published five single-author papers on the spectroscopy of diamonds and rubies. For a young researcher, this was an extraordinary achievement. Many scientists go their entire careers without producing five papers; Anna did it in three years, while also assisting with other laboratory work and pursuing her own research interests.

Her papers appeared in the *Proceedings of the Indian Academy of Sciences*, a prestigious journal founded by Raman himself. Each paper represented months of painstaking work—long exposures, careful measurements, detailed analysis.

Anna Mani’s Early Papers (1942-1945)		
Year	Title	Focus
1942	Fluorescence and absorption of light in Ruby	Ruby spectra
1944	The fluorescence and absorption spectra of diamond in the visible region	Diamond spectra
1944	Polarisation of Raman scattering and of fluorescence in diamond	Polarisation effects
1944	Spectroscopic study of luminescence patterns in diamond	Luminescence patterns
1945	Excitation curves of luminescence in diamond	Excitation mechanisms

The most cited of these papers, “The fluorescence and absorption spectra of diamond in the visible region,” reported a second set of fluorescence peaks in diamonds. This was a significant finding—it showed that diamonds had more complex optical properties than previously understood, and it provided evidence that could be used to test competing theories of diamond structure.

What made Anna’s work particularly valuable was its thoroughness. She

didn't just measure spectra; she studied how they changed with temperature, with orientation, with different excitation conditions. She recorded the polarisation of emitted light—whether it vibrated in one direction or another—providing additional information about the crystal structure. She measured excitation curves, showing which wavelengths of light were most effective at producing fluorescence.

All of this required not just technical skill but intellectual creativity. Anna had to design experiments that would answer specific questions, interpret complex data, and draw conclusions that were justified by the evidence. She was not simply following Raman's instructions; she was thinking independently, making discoveries, and contributing new knowledge.

The diamonds Anna studied came from various sources. Some were from Raman's personal collection; others were borrowed from museums or private owners. Each diamond had its own history, its own character. Anna learned to recognise individual diamonds by their spectra, to identify subtle differences that distinguished one stone from another.

This intimate knowledge of her materials gave Anna insights that less dedicated researchers might miss. She noticed, for example, that diamonds with different colours had different fluorescence patterns. Colourless diamonds behaved one way; yellow diamonds another; rare blue diamonds yet another. These differences reflected variations in impurities and crystal structure, and they provided clues to the conditions under which the diamonds formed.

Anna's work contributed to a broader understanding of diamonds that would later have practical applications. Today, spectroscopy is used to identify synthetic diamonds, to detect treatments that enhance colour, and to determine the geographic origin of gemstones. The techniques Anna helped develop are now standard in the gemological industry.

Five Papers, No Degree

In 1945, Anna submitted her Ph.D. dissertation to the University of Madras. The dissertation summarised her five years of research, her five published papers, and her contributions to understanding the spectroscopy of diamonds and rubies. It was, by any objective measure, a substantial and significant piece of scientific work.

The University of Madras rejected it!

The reason? Anna did not have a Master's degree!

When Anna joined IISc in 1940, she held only a bachelor's degree. The normal path to a Ph.D. would have been to complete a master's first,

then begin doctoral research. But Anna's brilliance and Raman's willingness to accept her had bypassed this requirement. She had been admitted directly to the research programme, working alongside students who already had their master's degrees.

Now, when it came time to award the Ph.D., the university's rules intervened. The regulations clearly stated that a candidate for the Ph.D. must hold a master's degree. Anna did not. Therefore, no Ph.D.

Rules and Exceptions

Science depends on rules. Experiments must follow protocols; data must be analysed according to accepted methods; conclusions must be justified by evidence. Without rules, science would be chaos, with no way to distinguish valid results from wishful thinking.

But rules can also be obstacles. When they are applied rigidly, without regard for individual circumstances, they can prevent worthy candidates from receiving recognition. The rule requiring a master's degree for a Ph.D.

What Happened to Anna's Dissertation?

Anna Mani's Ph.D. dissertation never earned her a degree, but it was not lost. It remains in the library of the Raman Research Institute in Bangalore, where it can be consulted by researchers interested in the history of Indian science.

The dissertation is a substantial document, running to several hundred pages. It contains detailed descriptions of Anna's experiments, her raw data, her analytical methods, and her conclusions. For historians of science, it provides a window into the workings of Raman's laboratory and the state of spectroscopy in 1940s India.

For readers today, the dissertation is a reminder of what Anna accomplished despite the obstacles she faced. The work exists; the knowledge is there; the contribution to science is real. The missing degree does not change any of that.

Perhaps most striking is Anna's own attitude. She never complained about the injustice, never sought to have the decision reversed, never let it affect her work. She simply continued doing science, finding other ways to contribute, other institutions to work with, other problems to solve. The rejection was a setback, but it was not the end.

made sense in general—it ensured that doctoral candidates had adequate preparation. But in Anna’s case, the rule ignored the reality that she had already demonstrated her preparation through five years of successful research.

The university could have made an exception. It could have recognised that Anna’s published papers were worth more than any master’s degree. It chose not to.

This was a profound injustice, and it left a permanent mark on Anna’s relationship with formal academia. For the rest of her life, she would work without the credential that normally signified the highest level of scientific training. She would never be “Dr. Mani”—only “Miss Mani” or “Mrs. Mani” (though she never married) or simply “Anna Mani.”

Yet she did not let this setback define her. When asked about it years later, she expressed no bitterness. She had done the work; she knew its value; she did not need a piece of paper to validate her achievements. The dissertation itself remained in the library of the Raman Research Institute, available for any curious reader to consult, a silent testament to work that deserved recognition but did not receive it. The denial of her Ph.D. might have discouraged a lesser spirit. For Anna, it was a turning point—but not the kind one might expect. Instead of abandoning science, she began to look for new directions, new fields where she could apply her skills.

In 1945, the same year her dissertation was rejected, Anna received another piece of news: she had been awarded a government scholarship for an internship in England. The scholarship was in meteorological instrumentation—not her chosen field, but an area where India desperately needed expertise.

Anna accepted. She would go to England, learn about weather instruments, and bring that knowledge back to India. It was not the path she had planned, but it would prove to be the path that led to her life’s work.

CHAPTER 3

From Diamonds to the Sky: A New Path

"Accurate measurements are the foundation of understanding nature."

- Anna Mani

A Troop Ship to England

The year 1945 was a momentous one in world history. In Europe, the Second World War was finally ending. In India, the demand for independence had reached a fever pitch. And for Anna Mani, it was the year she left everything familiar behind and sailed for England.



(Figure 3.1: Anna Mani with friends in the U.K. around 1945, Image Courtesy: Anna Philipose, Source: <https://annamani.co.in/education.html>)

Travel in 1945 was not the comfortable experience it is today. Commercial flights were rare and expensive; most people travelled by ship. And because the war had just ended, civilian travel was still affected by military priorities. Anna travelled on a troop ship—a vessel designed to carry soldiers, converted to carry passengers in the aftermath of war.

The ship was crowded and uncomfortable. Accommodations were basic, privacy minimal, and the voyage long—weeks at sea, crossing oceans, with nothing but water in every direction. For a woman who had spent her entire life in India, who had never travelled abroad, this must have been both exciting and terrifying.

The ship stopped at various ports along the way—Colombo, Aden, and Port Said. At each stop, Anna caught glimpses of places she had only read about. The sights, sounds, and smells of these foreign ports must have been

Try to Imagine

You are twenty-seven years old, and you have never left India. Now you are on a ship crossing the Indian Ocean, heading for a country you have only read about. The ship is crowded with soldiers returning home, with other students seeking education, with people whose languages you don't speak and whose customs you don't know.

You have no email, no phone, no way to contact home. Your family won't hear from you for weeks. Ahead lies England—cold, grey, recovering from war, utterly different from the tropical lushness of Kerala.

What do you feel? Excitement at the adventure? Fear of the unknown? Determination to succeed? All of these, probably. Anna Mani, as we have seen, was not one to be deterred by difficulty. She faced the voyage with the same quiet determination she had brought to her laboratory work.

overwhelming. But Anna, ever observant, absorbed everything, storing away impressions that would stay with her for life.

Finally, after weeks at sea, the ship reached England. Anna stepped onto British soil for the first time, carrying little more than her determination and her dreams. She was about to begin a new chapter in her life.

Learning the Language of Instruments

Anna's destination was London, specifically Imperial College, where she would study meteorological instrumentation. This was not her choice, the

scholarship was specifically for this field, but she approached it with her characteristic thoroughness.

Meteorology, she quickly learned, was not just about watching the sky. It was a science built on precise measurement. To predict the weather, you needed to know the temperature, pressure, humidity, wind speed, and a dozen other variables. And to measure those variables accurately, you needed instruments that were carefully designed, calibrated, and maintained.

At Imperial College, Anna immersed herself in this world. She learned how thermometers worked—not just in theory, but in practice, understanding the properties of the liquids they contained, the expansion rates of different materials, and the sources of error that could affect readings. She studied barometers, learning how atmospheric pressure could be measured using mercury columns or aneroid cells. She explored hygrometers, which measure humidity, and anemometers, which measure wind speed.

Think Like a Scientist: How Do Weather Instruments Work?		
Instrument	What It Measures	How It Works.
Thermometer	Temperature	Liquid (usually mercury or alcohol) expands when heated, rises in a calibrated tube.
Barometer	Atmospheric pressure	Mercury column rises or falls with pressure; aneroid barometers use a flexible metal cell.
Hygrometer	Humidity	A hair hygrometer uses human hair that lengthens with humidity; a psychrometer uses wet and dry bulbs.
Anemometer	Wind speed	Cups rotate faster in stronger wind; rotations are counted electronically or mechanically.
Rain gauge	Precipitation	Funnel collects rain in a graduated cylinder; tipping bucket records each 0.1 mm.
Sunshine recorder	Sunshine duration	The glass sphere focuses sunlight on paper card, burning a trace.

But Anna learned more than how individual instruments worked. She learned about calibration, the process of comparing an instrument's readings to known standards to ensure accuracy. She learned about standardisation, ensuring that instruments made in different places would give comparable readings. She learned about the entire infrastructure of measurement that underlay modern meteorology.

This was exactly the knowledge India needed. In 1945, India still depended entirely on imported meteorological instruments. When a thermometer broke or a barometer malfunctioned, it had to be sent abroad for repair. There was no indigenous manufacturing capability, no local expertise in instrument design, and no capacity to develop instruments suited to Indian conditions.

Anna realised that her training was not just about learning a new skill. It was about acquiring knowledge that could help make India self-reliant. Every lesson she learned, every technique she mastered, could be brought back and applied at home.

Walking the British Countryside

Anna's training was not confined to classrooms and laboratories. The British Meteorological Office, which sponsored her programme, believed in practical experience. So Anna travelled extensively across Britain, visiting meteorological stations of all kinds.

She went to Scotland, to the Eskdalemuir observatory, which housed magnetic and seismological instruments alongside weather equipment. She visited stations in the English countryside, on the coast, and in cities. She saw how instruments were deployed in different environments, how they were maintained, how data was collected and transmitted.

Mini-Profile: Eskdalemuir Observatory

Eskdalemuir Observatory, located in the hills of southern Scotland, was one of the most important meteorological stations in Britain when Anna visited. Established in 1908, it was sited in a remote area to avoid interference from human activity—essential for its magnetic measurements.

The observatory recorded not just weather but also earthquakes (using seismometers) and variations in the Earth's magnetic field. Its isolated location meant that staff lived on site, creating a close-knit community of scientists and technicians.

These visits were educational in multiple ways. Anna learned about the practical challenges of weather observation—the effects of exposure, the need for regular maintenance, and the importance of site selection. She also learned about the British landscape, about the people who worked at these stations, about the rhythms of rural and urban life.

For Anna, visiting Eskdalemuir was a glimpse into a different way of doing science—a long-term commitment to continuous observation, generation after generation, building datasets that would outlast any individual researcher. She would later apply this lesson in India, establishing monitoring networks designed to operate for decades.

Anna was also an enthusiastic walker. Whenever she had free time, she would explore the countryside on foot, observing the landscape, the vegetation, and the wildlife. This was not just recreation; it was a continuation of the habits she had developed as a child in Travancore. She was always observing, always learning, always connecting what she saw with what she knew.

Her letters from this period, preserved in archives, reveal a woman who was thoroughly enjoying her British adventure. She wrote about the green English countryside, about the friendliness of the people she met, and about the peculiarities of British food and customs. She also wrote about her work—the instruments she was studying, the techniques she was learning, the plans she was making for her return to India.

The National Physical Laboratory

A particularly important part of Anna's training was a three-month stint at the National Physical Laboratory (NPL) in Teddington, near London.

The NPL was Britain's national standards laboratory—the place where the fundamental units of measurement were defined and maintained. Here, scientists worked to ensure that a kilogram in London was the same as a kilogram in Bombay, that a metre in Manchester matched a metre in Madras, that measurements made anywhere in the British Empire (and later, the Commonwealth) could be compared with confidence.

For meteorological instruments, standards are particularly important. Weather data from different stations must be comparable to be useful. If one thermometer reads 30°C when it's really 29°C, that error could affect forecasts, climate records, and scientific research.

At the NPL, Anna learned about the painstaking work of maintaining standards. She saw how mercury barometers were calibrated against primary standards, how thermometers were compared in controlled baths, and how

Think Like a Scientist: Why Standards Matter?

Imagine if every ruler were a different length. A carpenter in one city might cut wood that wouldn't fit in another city. A scientist measuring temperature in Pune might get readings that couldn't be compared with measurements in Delhi. Chaos would result.

That's why standards are essential. They ensure that measurements are consistent across time and space. The International System of Units (SI) defines seven base units—metre, kilogram, second, ampere, kelvin, mole, candela—from which all other units are derived. National laboratories like the NPL maintain standards that realise these definitions with the highest possible accuracy.

instruments were tested for accuracy and stability. She learned that good measurements require not just good instruments, but a whole infrastructure of calibration and quality control.

This knowledge would prove invaluable when Anna returned to India and began building her own calibration facilities. She understood that self-reliance meant more than manufacturing instruments; it meant establishing the standards against which those instruments could be verified. Without standards, manufactured instruments might be inaccurate, inconsistent, or unreliable.

Return to a Free India

Anna's three years in England coincided with one of the most momentous events in Indian history: independence.

On August 15, 1947, while Anna was still in Britain, India became a free nation. The British flag was lowered; the tricolour was raised. Jawaharlal Nehru delivered his famous "tryst with destiny" speech. Millions celebrated in the streets.

Anna followed these events from afar, reading newspapers, listening to radio broadcasts, and discussing with fellow Indian students. The news filled her with joy—and with a renewed sense of purpose. Independent India would need scientists, engineers, and technicians to build the nation. She could contribute.

In 1948, Anna completed her training and prepared to return home. She had learned everything she could about meteorological instrumentation—the theory, the practice, the standards, the manufacturing. She had visited dozens

of stations, met hundreds of people, and absorbed knowledge that was not available in any textbook. Now it was time to put that knowledge to use.

The voyage back to India was very different from the outward journey. Anna was no longer a nervous student heading into the unknown; she was an expert returning to contribute to her nation. The troop ship that had carried her to England was replaced by a more comfortable passenger vessel. The uncertainty about the future was replaced by a clear purpose.

As the ship approached Bombay harbour, Anna stood on deck, watching the Indian coastline emerge from the sea. She had been away for three years. She had learned much. Now she was home.

And the real work was about to begin.

CHAPTER 4

Building a Self-Reliant India: The Pune Years - Entering into a New World of Meteorology

"Wrong measurements are worse than no measurement"

- Anna Mani

Joining the India Meteorological Department

In 1948, Anna Mani joined the India Meteorological Department (IMD) at Pune. She was thirty years old, newly returned from England, and ready to apply everything she had learned.



(Figure 4.1: India Meteorological Department, Pune,
Image Source: <https://www.imdpune.gov.in/history.php>)

Mini-Profile: S.P. Venkiteshwaran

S.P. Venkiteshwaran was a scientist with a dream. He wanted India to manufacture its own weather instruments, to stop depending on imports from Britain and other countries. This was not just a matter of pride; it was a matter of practicality. Instruments designed for British conditions might not work optimally in India. Instruments that had to be sent abroad for repair could be out of service for months. A self-reliant India needed its own instruments.

Venkiteshwaran was also a nationalist. He believed that scientific independence was as important as political independence. By building Indian instruments, Indian scientists could demonstrate their capability, reduce dependence on former colonial powers, and create infrastructure that would serve the nation for generations.

The IMD in 1948 was a department in transition. During the colonial period, it had focused primarily on serving British interests—forecasting for shipping, tracking monsoon patterns that affected agriculture, and collecting data that might be useful for imperial administration. Indian scientists worked in the department, but leadership positions were mostly held by Britons.

With independence, everything changed. Indian scientists now led the department. Indian priorities shaped its work. And there was a new emphasis on building indigenous capability—on making India self-reliant in meteorology, rather than dependent on foreign expertise and equipment.

Anna was assigned to the Instruments Division, headed by S.P. Venkiteshwaran, a visionary scientist with tremendous energy and a clear mission: to make India self-sufficient in meteorological instruments.

Anna found in Venkiteshwaran a kindred spirit. Both were committed to self-reliance. Both believed in the importance of high-quality work. Both were willing to put in the long hours and hard effort required to build something from nothing.

Before 1947, even the simplest meteorological instruments—thermometers, barometers, rain gauges—were imported. Indian workshops did little more than repair damaged equipment. There was no indigenous manufacturing capability, no design expertise, no quality control infrastructure.

Venkiteshwaran had already begun changing this. He had secured funding for a modern instrument manufacturing facility, complete with

testing and calibration laboratories. He had imported precision machinery from Europe. He had begun recruiting scientists and technicians who could make his vision real.

Anna joined this effort at exactly the right moment. Her training in England—her knowledge of instrument design, calibration, and standards—made her the ideal person to lead the technical work. Her determination and work ethic meant she would not rest until the job was done.

The Vision of S.P. Venkiteshwaran

Venkiteshwaran's vision was ambitious. He wanted to build a facility where instruments could be designed, manufactured, tested, calibrated, and deployed—all under one roof. Raw materials would enter one end of the workshop; finished instruments would emerge from the other, ready for use across India.

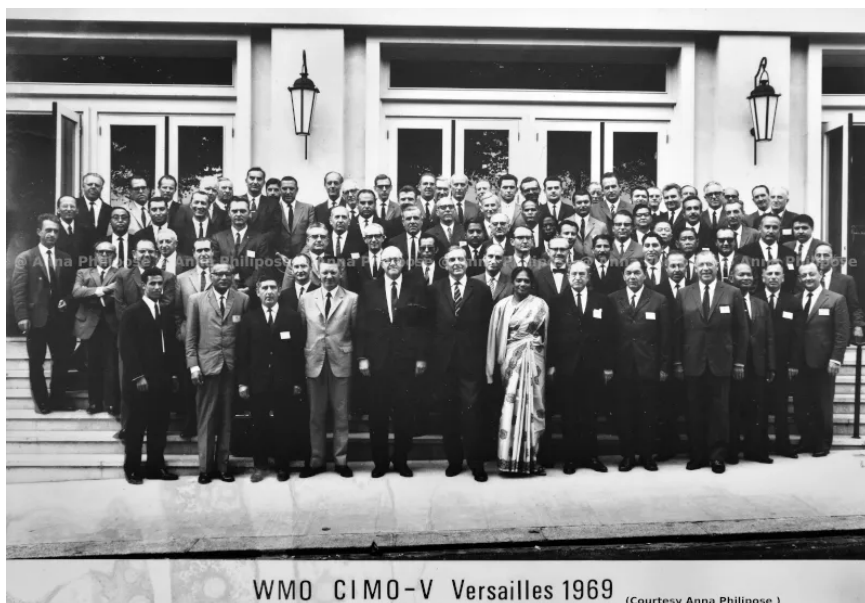
This was not the way things were usually done. In most countries, instrument manufacturing was distributed across multiple companies, each specialising in particular products. Calibration was handled by separate laboratories. Design was done by engineers who might never see the instruments being used.

But Venkiteshwaran understood that India's situation was different. There was no existing instrument industry to build upon. There were no specialised companies with decades of experience. If India wanted self-reliance, it would have to build capability from scratch—and that meant concentrating expertise in a single facility.

Think Like an Engineer: The Workshop Concept

Venkiteshwaran's workshop was designed to be self-contained. It included:

- **Foundry:** For casting metal parts
- **Milling shop:** For shaping metal precisely
- **Turning shop:** For creating cylindrical parts
- **Jig boring shop:** For creating precision holes
- **Sheet metal shop:** For forming enclosures and mounts
- **Plating shop:** For applying protective coatings
- **Painting shop:** For finishing
- **Carpentry shop:** For wooden parts and packaging
- **Packing shop:** For preparing instruments for shipment



(Figure 4.2: Anna Mani at WMO CIMO-V Versailles 1969, Image Courtesy: Anna Philipose, Source: <https://annamani.co.in/conferences.html>)

This was not just a collection of shops; it was an integrated system. A design could move from drawing board to finished product without ever leaving the facility. Every step could be controlled, every quality standard enforced, every problem identified and fixed immediately.

The machinery in Venkiteshwaran's workshop was world-class—much of it imported from Europe. The jig boring machines, which could drill holes with microscopic precision, were so good that industries in Bombay and

Mini-Profile: K.R. Ramanathan

K.R. Ramanathan (1893-1984) was one of the pioneers of Indian meteorology. After studying under C.V. Raman (no relation), he joined the IMD and eventually became its director. He later founded the Physical Research Laboratory in Ahmedabad with Vikram Sarabhai.

Ramanathan's contributions to atmospheric science were immense. He studied the monsoon, atmospheric radiation, and ozone. He served as president of the International Association of Meteorology and Atmospheric Physics and helped plan the International Geophysical Year.

Pune would send their own work to be done at the IMD facility. Here was a government workshop that private companies envied.

But machinery alone was not enough. Trained manpower was scarce in post-independence India. There were a few industrial training institutes, and even fewer people with experience operating precision machinery. Anna and Venkiteshwaran had to build their team from scratch—recruiting promising individuals, training them, mentoring them, and building a culture of excellence.

A Workshop Like No Other

Anna threw herself into this work with characteristic energy. Her first task was to supervise the construction of a recording rain gauge, an instrument that would automatically record rainfall, rather than just collecting it for manual measurement.



(Figure 4.3: Rain gauge @IMD Pune, Image Source: <https://annamani.co.in/instruments.html>)

This might sound simple, but it was not. A recording rain gauge must accurately measure rainfall, record it over time, and operate reliably in all weather conditions. The mechanism must be sensitive enough to detect small amounts of rain, robust enough to withstand storms, and precise enough to produce useful data.

The first rain gauge was a success. Encouraged, Anna moved on to more complex instruments: hygrographs for recording humidity, thermographs for continuous temperature records, barographs for pressure, and anemographs for wind. Each instrument presented its own challenges, its own design problems, its own manufacturing difficulties.

Recording vs. Indicating Instruments

Meteorological instruments come in two basic types:

Indicating instruments show the current value of a variable. A mercury thermometer indicates the current temperature. A barometer indicates the current pressure. You read them at a particular moment, and you get a number.

Recording instruments (often called «graphs») produce a continuous record over time. A thermograph uses a pen to draw a line on a rotating drum, creating a permanent record of temperature variations. A barograph does the same for pressure.



(Figure 4.4: Anna Mani in Payeme, Switzerland, 1956.
Image Source: World Meteorological Organization)

Recording instruments are more complex than indicating instruments. They require mechanisms to move paper, pens that write reliably, and careful calibration to ensure the record accurately represents reality. They also produce vastly more information—not just the temperature now, but how it changed over hours and days.

Anna's work on recording instruments was essential for building a modern meteorological service. With recording instruments, India could begin to understand not just the state of the atmosphere at particular moments, but its continuous evolution over time.

By 1953, just five years after Anna joined the IMD, the Instruments Division had stopped importing all surface instruments. Every thermometer, barometer, hygrometer, and rain gauge used by the IMD was now manufactured in Pune. This was an extraordinary achievement, a complete transformation of Indian meteorological capability in half a decade.

Standardising the Instruments

Manufacturing instruments was one thing; ensuring they were all accurate and consistent was another. Anna's work on standards was as important as her work on manufacturing.

She began by establishing primary standards for pressure, temperature, and wind measurement. Pressure was particularly critical—accurate pressure measurements were fundamental to weather forecasting. Anna travelled across India, setting up regional standard barometers and comparing them with the primary standards kept at Pune and Kolkata.

Think Like a Scientist: Traceability

When you use a thermometer, how do you know it's accurate? Because it was compared with a standard. And that standard was compared with a higher standard. And so on, up to the primary standard that defines the unit itself.

This chain of comparisons is called traceability. It ensures that measurements made anywhere can be related to the fundamental definitions of units. A thermometer in Pune is accurate because it was calibrated against a standard in Pune. That standard was calibrated against a national standard in Delhi. That national standard was calibrated against international standards maintained by the International Bureau of Weights and Measures.

Anna built this traceability chain for Indian meteorology. She established primary standards at Pune and Kolkata, then created secondary standards that could be taken to regional centres, and then ensured that every instrument in the field was calibrated against those standards.

For each type of instrument, Anna prepared detailed engineering specifications and technical manuals. These documents described exactly how the instrument should be designed, what materials should be used, how it should be assembled, how it should be tested, and how it should be calibrated. They ensured that instruments made years apart, by different technicians, would be identical in performance.

In total, Anna standardised nearly a hundred different weather instruments. Each one had its own specifications, its own drawings, its own manufacturing process. Each one represented hours of work—analysing existing designs, adapting them for Indian conditions, documenting every detail.

Leading 121 Men

By 1953, Anna Mani was in charge of the Instruments Division. She led a team of 121 men—scientists, engineers, technicians, and support staff.

This was unusual. In the 1950s India, women in leadership positions were rare. Women leading teams of men were almost unheard of. Anna's position was a testament to her expertise, her dedication, and the respect she had earned from colleagues.

But it was not always easy. Some of the men under her supervision initially resented taking orders from a woman. Others doubted whether a woman could understand the technical details of instrument manufacturing. A few actively tried to undermine her authority.

Anna's response was consistent: she focused on the work. She demonstrated her knowledge in every interaction. She held herself to the highest standards, expecting no less from others. She was fair but firm, approachable but professional.

“Wrong measurements are worse than no measurements,” she would tell her team. This became a guiding principle for the division. It wasn't enough to produce instruments; they had to be accurate. It wasn't enough to collect data; it had to be reliable. Quality was not negotiable.

Anna's leadership style was distinctive. She led by example—working long hours, mastering every detail, never asking others to do what she wouldn't do herself. She was demanding but fair, expecting excellence but recognising



*(Figure 4.5: Anna Mani at the International Radiation Symposium, Sendai, Japan, 1972.
Image Courtesy: Anna Philipose, Source: <https://annamani.co.in/conferences.html>)*

achievement. She built a team that was loyal, dedicated, and proud of its work.

Her colleagues remembered her as “terrifying” if you made mistakes, but also as deeply caring. When junior colleagues were transferred to Delhi, she made sure they had warm clothing, something they might not have needed in Pune. When someone was struggling with a problem, she would stay late to help. When a colleague showed interest in something outside work—like writing—she encouraged it.

These personal touches made the Instruments Division more than a workplace. It became a community of people committed to a common purpose: building India’s meteorological capability.

The Metric System Challenge

In the 1950s, India made a momentous decision: to adopt the metric system of measurement. Instead of inches, feet, and pounds, India would use metres, kilograms, and litres. Instead of Fahrenheit, India would use Celsius.

This was a sensible decision—the metric system was (and is) the international standard, used by scientists worldwide. But for the IMD, it created a massive challenge.

Every instrument in use—every thermometer, every barometer, every rain gauge—was calibrated in British units. Every instruction manual assumed British units. Every technician was trained to work with British units. Converting to metric meant replacing or recalibrating every instrument, rewriting every manual, and retraining every technician.

Anna led this effort. It was, in her own words, “one of the most difficult tasks” of her career.

The first step was to revise all engineering specifications and drawings. Every dimension had to be converted from inches to millimetres. Every tolerance had to be recalculated. Every part had to be redesigned to metric specifications.

The workshop had to be re-equipped. Machines calibrated in inches needed new scales or new parts. Tools designed for British fasteners needed replacement. Suppliers had to be found for metric materials. Adding to the task was the field network. Hundreds of observation stations across India had to be visited, their instruments replaced or recalibrated. Data collection had to continue during the transition, which meant maintaining two parallel systems until the conversion was complete.

It took years, but Anna’s team succeeded. By the early 1960s, India’s meteorological network was fully metric, one of the first in the world to make the transition. This was a legacy that would serve the country for generations.

CHAPTER 5

Measuring the Invisible: Ozone and the Sky

“One test is worth ten thousand opinions”.

- Anna Mani

The Mysterious Layer Above

High in the atmosphere, about 15 to 35 kilometres above the Earth’s surface, lies a layer of gas that makes life possible. This is the ozone layer, a region where ozone molecules (O_3) are concentrated, absorbing most of the Sun’s harmful ultraviolet radiation.

Without the ozone layer, life as we know it could not exist. Ultraviolet radiation would damage DNA, cause cancers, and disrupt ecosystems. The fact that we can walk in sunlight without immediate harm is thanks to this thin, invisible layer far above our heads.

In 1960, when Anna Mani began studying ozone, much of this was not yet understood. Scientists knew that ozone existed in the atmosphere, but its distribution, its variations, and its role were still being explored. The word “ozone” was not famous; it was just another topic for scientific investigation.

In 1960, Anna Mani started her work on measuring atmospheric ozone, and then later in 1963, at the request of Vikram Sarabhai, she set up a meteorological observatory and an instrumentation tower at the Thumba rocket launching facility. Anna’s interest in ozone was sparked by K.R. Ramanathan, a legendary figure in Indian meteorology. Ramanathan had retired from the IMD and became director of the Physical Research Laboratory in Ahmedabad, but he remained deeply involved in atmospheric science. As president of the International Ozone Commission, he understood the importance of ozone research, and he understood that India could contribute.

Breakout Box: What is Ozone?

Ozone (O_3) is a molecule made of three oxygen atoms. Ordinary oxygen, which we breathe, is O_2 , i.e., two oxygen atoms. The extra atom makes ozone chemically reactive and gives it distinctive properties.

In the stratosphere (15-35 km up), ozone is formed when ultraviolet light splits ordinary oxygen molecules, and the free atoms combine with other oxygen molecules. This process also absorbs ultraviolet radiation, protecting life below.

Near the ground, ozone is a pollutant, a component of smog that can damage lungs and plants. The same molecule that protects us high above can harm us close to Earth. Context matters.

Understanding ozone means understanding both where it is and how it varies. This requires measurements—lots of them, over long periods, at many locations. Anna Mani recognised this and made ozone measurement one of her priorities.

For Anna Mani, Ramanathan was both mentor and inspiration. It was he who urged her to take up ozone research, who helped secure equipment, and who guided her through the complexities of a new field. When Anna later received the INSA K.R. Ramanathan Medal, it was a fitting tribute to their relationship.

K.R. Ramanathan's Inspiration

Ramanathan had acquired a Dobson spectrophotometer for the Physical Research Laboratory, an instrument for measuring total ozone in a column of atmosphere. He had also helped the IMD establish a network of four Dobson instruments at locations across India.

But measuring total ozone was only part of the story. To understand ozone fully, one also needed to know its vertical distribution—how much ozone is at each altitude. This required a different instrument: an ozonesonde.

An ozonesonde is a device carried aloft by a balloon, measuring ozone concentration as it rises. It transmits data back to ground stations, building a profile of ozone from the surface up to 35 kilometres or more.

In 1957, a British scientist named A.W. Brewer had developed an electrochemical ozonesonde at Oxford University. It was considered the most reliable design available. Ramanathan suggested that Anna develop a similar instrument in Pune. Anna took up the challenge in 1962.

Think Like a Scientist: Electrochemical Ozone Detection

The chemistry behind the ozonesonde is elegant:

1. Air is bubbled through potassium iodide (KI) solution.
2. Ozone (O_3) reacts with iodide ions (I^-) to produce iodine (I_2).
3. Each ozone molecule produces one iodine molecule.
4. The iodine reacts at an electrode, producing an electrical current.
5. The current is proportional to the ozone concentration.

This reaction is specific to ozone—other gases don't interfere—and sensitive enough to detect the tiny ozone concentrations found in the atmosphere (typically a few parts per million).

But the practical implementation required solving many problems: designing a pump that would work at low pressures, ensuring the solution didn't freeze at high altitudes, protecting the electronics from temperature extremes, and transmitting the weak signal reliably over distance.

Building the Indian Ozonesonde

Developing an ozonesonde was not easy. The instrument had to be light enough to be carried by a balloon, robust enough to survive the ascent, and sensitive enough to measure tiny concentrations of ozone. It had to transmit data reliably over long distances, and it had to operate in the harsh conditions of the upper atmosphere.

The principle was simple but the implementation was complex. As the balloon rose, a small pump would draw in ambient air and bubble it through a potassium iodide solution. Ozone in the air would react with the solution, liberating iodine atoms. Each ozone molecule would release two iodine atoms, and each iodine atom would transport two electrons across a sensor. This tiny electrical signal would modulate a radio transmitter, sending data to ground stations.

Anna was assisted initially by E.L. Simmons, a physics lecturer from Wilson College in Mumbai who happened to be a former student of A.W. Brewer at Oxford. (In 1957, British scientist A.W. Brewer had developed an electrochemical ozonesonde at Oxford University). Simmons understood the instrument's intricacies and could guide Anna through the challenges.



(Figure 5.1: Anna Mani and a colleague work on a radiosonde, a balloon-borne weather-measuring equipment. Photo credit: World Meteorological Organization.)

Despite their expertise, the project faced repeated failures. Balloons burst prematurely. Pumps malfunctioned. Electronics failed. Data was lost. Months of work would be destroyed in minutes.

But Anna persisted. Each failure taught her something. Each problem, once solved, wouldn't recur. Slowly, steadily, the ozonesonde improved.

In September 1964, after two years of work, Anna's team produced the first successful result from Pune. The balloon rose, the instruments worked, and the data came through. India had its own ozonesonde.



(Figure 5.2: Anna Mani with international scientists in Tübingen, Germany, in 1965 - © dpa, Image Source: Internet)

Ongoing Success

The success was just the beginning. Over the next five years, Anna and her team—particularly C.R. Sreedharan and G. Venugopal - refined the instrument, improving its reliability and accuracy. Systematic soundings began from Thiruvananthapuram, Pune, and New Delhi.

In 1970, the Indian ozonesonde participated in an international comparison at Hohenpeissenberg in Germany. Scientists from around the world brought their instruments, launched them together, and compared results. The Indian sonde performed excellently, as good as any in the world.

This was a moment of pride. India, which had once imported all its meteorological instruments, was now exporting expertise. The ozonesonde developed in Pune was recognised as world-class.

The ozonesonde became Sreedharan's life's work. He continued to improve it, to conduct soundings, to analyse data. Decades later, he could look back on a career built on the foundation Anna had laid.

With reliable soundings now possible, India began to build a picture of ozone over the tropics. The data revealed patterns- seasonal variations, links to weather systems, differences between locations. India was contributing to global understanding of the atmosphere.

Mini-Profile: C.R. Sreedharan

C.R. Sreedharan was one of Anna Mani's closest collaborators and the scientist who, with G. Venugopal, took the ozonesonde project to its full potential. He later wrote Anna's biographical memoir for the Indian National Science Academy.

Sreedharan remembered Anna as a demanding but supportive mentor. She expected excellence but provided the resources and guidance needed to achieve it. She gave her junior colleagues responsibility and trusted them to deliver. She celebrated their successes and stood by them during failures.

Detecting the Ozone Hole

In the 1970s, scientists began to worry about threats to the ozone layer. **Mario Molina** and **F. Sherwood Rowland** proposed that chlorofluorocarbons (CFCs)—chemicals used in refrigerators, air conditioners, and aerosol sprays—could destroy ozone. Their theory was controversial, and CFC manufacturers fought back fiercely.

Then, in 1985, a British scientist named Joe Farman published a shocking discovery: the ozone layer over Antarctica was thinning dramatically each spring. An «ozone hole» had formed. Mario Molina and F. Sherwood Rowland were vindicated, and they later won the Nobel Prize in Chemistry in 1995. But here's the remarkable thing: Indian scientists had detected the ozone hole earlier, in 1982.

From 1982 onward, India's scientific expeditions to Antarctica included ozone soundings using the instrument Anna Mani had developed. These soundings clearly showed the dramatic ozone depletion over the continent during September and October. Indian data provided independent confirmation of what Farman would later publish. India's early detection of the ozone hole was a testament to Anna Mani's foresight. She had started ozone work in 1960, years before anyone understood the urgency. She had built an instrument that could measure ozone reliably. She had trained a team that could conduct soundings in extreme conditions. When the ozone hole became a global concern, India was ready.

International Recognition

Anna's work on ozone brought her international recognition. She became a member of the International Ozone Commission, joining the world's

Think Like a Scientist: Why Antarctica?

Why does the ozone hole form over Antarctica and not elsewhere? Several factors combine:

1. **Extreme cold:** The Antarctic stratosphere gets very cold in winter, forming polar stratospheric clouds.
2. **Cloud chemistry:** Chemical reactions on cloud particles convert chlorine from CFCs into forms that destroy ozone.
3. **Isolation:** The polar vortex isolates Antarctic air, preventing mixing with mid-latitude air.
4. **Sunlight:** When sunlight returns in spring, it triggers the ozone-destroying reactions.

The result is a rapid, dramatic loss of ozone—a hole that forms each spring and recovers later in the year. Without the special conditions over Antarctica, similar depletion doesn't occur elsewhere (though global ozone has still decreased).

leading experts on atmospheric ozone. She participated in international meetings, contributed to global assessments, and helped shape the scientific understanding that would eventually lead to the Montreal Protocol, the international treaty that phased out CFCs.

The Montreal Protocol, signed in 1987, is one of the most successful environmental treaties in history. By phasing out ozone-depleting substances,

Breakout Box: The Montreal Protocol

The Montreal Protocol on Substances that Deplete the Ozone Layer was signed in 1987 and has been ratified by all UN member countries. It:

- Phased out production of CFCs and other ozone-depleting substances
- Provided developing countries with technical and financial assistance
- Established monitoring and reporting requirements
- Has been amended several times to strengthen controls

As a result of the protocol, atmospheric concentrations of ozone-depleting substances have declined, and the ozone layer is expected to recover by mid-century. The Montreal Protocol is often cited as a model for international environmental cooperation.

it has allowed the ozone layer to begin recovering. Without the work of scientists like Anna Mani, who measured ozone, tracked its variations, and documented its decline, the treaty would not have been possible.

Anna's contributions were recognised with the INSA K.R. Ramanathan Medal in 1987, awarded for outstanding contributions to atmospheric sciences. It was a fitting honour, linking her with the mentor who had inspired her ozone work.

Anna Mani's work contributed to the scientific understanding that made the Montreal protocol possible. Every time we hear that the ozone hole is healing, we are witnessing the fruits of decades of patient measurement-measurement that Anna helped make possible.

CHAPTER 6

The Sun and the Wind: A Vision for Renewable Energy

“I am very lucky because work in my professional capacity has been a hobby and joy to me.”

- Anna Mani

Solar Radiation: An Untapped Treasure

India is a tropical country. For most of the year, the sun beats down with an intensity that can be uncomfortable, even dangerous. But that same intensity represents an enormous resource-energy that could be harnessed to power homes, industries, and farms.

In the 1950s, when Anna Mani began thinking about solar energy, the idea of harvesting sunlight was still novel. Solar panels were expensive and inefficient. Large-scale solar power seemed like science fiction. But Anna, with her characteristic foresight, recognised that understanding solar radiation was the first step toward using it.

What is Solar Radiation?

Solar radiation is the energy emitted by the Sun that reaches Earth. It comes in several forms:

- **Direct radiation:** Sunlight that reaches the ground directly, without scattering
- **Diffuse radiation:** Sunlight that has been scattered by clouds, dust, or air molecules
- **Global radiation:** The total of direct and diffuse radiation on a horizontal surface
- **Net radiation:** The difference between incoming and outgoing radiation

Each type matters for different applications. Solar panels can use both direct and diffuse radiation, but their efficiency depends on the total energy

received. Building design might consider direct radiation for heating and diffuse radiation for daylighting. Agriculture depends on both.

Understanding solar energy potential requires extensive measurement, taken over long periods and across many locations.

Anna set out to provide those measurements.

Anna's work on solar radiation began during the International Geophysical Year (IGY) of 1957-58. The IGY was a global scientific effort, involving scientists from 67 countries, to study the Earth and its atmosphere. One of its many programmes was the measurement of solar radiation.

At the start of the IGY, India had no indigenous capability for measuring solar radiation. Instruments were imported, mostly from Europe. Only four radiation stations were operating.

Anna changed this.



(Figure 6.1: Anna Mani at an event, Source: RRI Digital Repository)

The International Geophysical Year

The International Geophysical Year was a landmark event in the history of science. For 18 months (July 1957 to December 1958), scientists around the world coordinated their efforts to study the Earth—its atmosphere, its oceans, its ice, its magnetic field, its connections to the Sun. India participated actively. The IGY provided an opportunity to build scientific

The International Geophysical Year (IGY) July 1957 to December 1958

The IGY was inspired by the International Polar Years of 1882-83 and 1932-33, but expanded to cover the entire Earth. Major achievements included:

- Discovery of the Van Allen radiation belts around Earth
- First measurements of ocean currents worldwide
- Extensive studies of Antarctica
- Launch of the first artificial satellites (Sputnik)
- Establishment of world data centres for geophysical data

The IGY set new standards for international scientific cooperation and demonstrated what could be achieved when nations worked together.

capability, to collaborate with international partners, and to contribute to global understanding. For Anna Mani, it was a chance to establish solar radiation measurement on a firm footing.

Anna's challenge was to build a network of radiation stations across India, a network that would continue operating long after the IGY ended. This required instruments, trained personnel, and rigorous procedures.

Measuring the “Anthropocene” even before the term was coined

Anna Mani was a pioneering Indian scientist known for her work in meteorology and renewable energy, especially solar and wind measurements. Long before the term ‘climate change’ became a global concern, she was already building the tools to measure it. While she did not directly use or develop the concept of the **Anthropocene**, her work is closely connected to the kinds of issues the Anthropocene highlights.

The Anthropocene Epoch is an unofficial geological term used to describe the most recent period in Earth's history in which human activities have begun to strongly shape Earth's climate and ecosystems. The word comes from the Greek *anthropos*, meaning “human,” and *-cene*, meaning “new,” and it was coined and popularized in 2000 by Eugene Stomer and Paul Crutzen.

The Anthropocene focuses on how human activity, especially industrialization and fossil fuel use, has altered Earth's climate systems. Anna Mani's research, particularly at the India Meteorological Department, helped to improve measurement of solar radiation, ozone, and wind energy potential. These are crucial for understanding climate patterns and for developing alternatives to fossil fuels. In that sense, Anna Mani's contributions support efforts to respond to Anthropocene challenges: better climate data, cleaner energy, and more informed environmental policy. So while she wasn't part of the Anthropocene debate itself, her scientific legacy plays an important role in how we study and address human impact on the planet.

Although she never used the term Anthropocene, which was coined decades later, her meticulous observations, calculations, and field studies effectively documented its early signs in India, capturing human influence on the atmosphere long before it had a name. Before carbon accounting became routine and climate justice entered global debate, Anna Mani had already done what mattered most: she had measured.

In doing so, she offered India its first clear glimpse of the era we now call the Anthropocene.

Initially, India participated with imported radiometers. But Anna was already planning for self-reliance. While operating the imported instruments, she began developing indigenous versions.

The first Indian instrument to emerge from the Pune laboratories was the pyrliometer, an instrument for measuring direct solar radiation. It was followed by the pyranometer (for global radiation) and the pyradiometer (for net radiation).

Building a National Network

Building a national radiation network was a massive undertaking. Stations had to be established at locations representing India's diverse climates—from the humid coasts to the arid deserts, from the tropical south to the Himalayan foothills.

Each station needed instruments, trained observers, and procedures for data collection and quality control. Instruments had to be calibrated regularly against standards. Data had to be checked for errors, compiled, and published.

Anna led this effort with her characteristic thoroughness. She and her colleague Oommen Chacko developed detailed procedures for instrument

operation, calibration, and data evaluation. They trained field workers to carry out measurements and maintain instruments. They established a system for scrutinising data at Pune, ensuring that only high-quality measurements were accepted.

By 1969, when Anna moved to Delhi as Deputy Director General, the Indian radiation network had grown to 24 stations, far exceeding the original IGY target. These stations were producing high-quality data that attracted international attention.

The quality of Indian radiation data was recognised by the World Meteorological Organization (WMO), which designated Pune as a Regional Radiation Centre for Asia. This meant that Pune would serve as a reference

Mini-Profile: Ommen Chacko

Ommen Chacko was Anna Mani’s close collaborator on solar radiation work. Together, they developed instruments, established networks, and published numerous papers. Chacko was responsible for much of the day-to-day work of training field staff, calibrating instruments, and processing data.

Their partnership exemplified Anna’s approach to science: she provided vision and leadership, but she also valued the contributions of her colleagues. Chacko’s expertise and dedication were essential to the success of the radiation programme.

for other countries in the region, helping them calibrate their instruments and standardise their measurements.

Anna was appointed chair of the WMO’s Working Group on Radiation Instruments and Observations, a position she held for two terms. In this role, she organised international comparisons of radiation instruments, coordinated surveys of radiation activities in Asia, and helped establish global standards for radiation measurement.

From Pyranometers to Data

Measuring solar radiation is one thing; turning those measurements into useful information is another. Anna understood that raw data was not enough—it had to be analysed, interpreted, and made accessible to users.

Over two decades, Anna and her colleagues amassed a vast database of solar radiation measurements from across India. They studied how radiation

varied with location, season, and weather conditions. They analysed the effects of clouds, dust, and water vapour. They developed methods for estimating radiation at locations without direct measurements, using sunshine records or cloud observations.

Estimating Solar Radiation

Direct measurement of solar radiation requires expensive instruments and trained observers. Many locations lack these. So, scientists have developed methods to estimate radiation from more readily available data.

One approach uses sunshine duration. If you know how many hours of bright sunshine a location receives, you can estimate how much solar radiation reaches the ground. The relationship isn't perfect- factors like cloud type, atmospheric pollution, etc., also play a role, but still, it's good enough for many purposes.

Another approach uses cloud observations. If you know how much of the sky is covered by clouds and what types of clouds, you can estimate their effect on radiation. This is more complex but can provide useful estimates.

Anna and her colleague S. Rangarajan developed both approaches for India, producing methods that could estimate radiation at hundreds of locations where only sunshine or cloud data were available.

Anna's work on solar radiation culminated in two monumental publications:

- **Handbook of Solar Radiation Data for India (1980):** Presented observed data from 18 stations
- **Solar Radiation over India (1981):** Presented computed data for 145 stations, using methods developed by Anna and Rangarajan

These handbooks became essential references for engineers, architects, and scientists working with solar energy. They provided the basic information needed to design solar systems, assess energy potential, and plan solar projects.

Handbooks That Changed Everything

The publication of Anna's solar radiation handbooks was a landmark event in Indian renewable energy. For the first time, anyone wanting to use solar energy in India had reliable data on what was available.

Solar water heaters could be designed for specific locations. Solar cookers could be tested against local conditions. Solar photovoltaic systems could be sized appropriately. Agricultural planners could assess the potential

for solar drying of crops. Architects could design buildings to maximise or minimise solar gain as appropriate.

Anna's vision extended beyond solar to wind energy. In the early 1980s, the Department of Non-Conventional Energy Sources (now the Ministry of New and Renewable Energy) asked her to assess India's wind energy potential.

Wind energy was even less developed than solar in India. There were few measurements of wind speeds at heights where turbines would operate, and little understanding of how wind varied across the country.

Anna took on the challenge with her usual energy. She organised a wind energy survey project, based in Bangalore, with a small team of dedicated workers. Their task: measure wind speeds at hundreds of locations across India, identify sites with good wind potential, and provide data for wind farm development.

This was a massive logistical challenge. Wind measurement masts, 20 metres tall, had to be erected at remote sites across the country. Instruments had to be selected, procured, calibrated, and deployed. Data had to be collected, checked, and analysed. Bureaucracies in different states had to be coordinated.

Wind Energy: The Final Frontier

Starting with 21 stations in 1984-85, mostly in Tamil Nadu and Gujarat, Anna's wind survey expanded rapidly. Anna Mani retired as the Deputy Director-General of the India Meteorological Department in 1976.

The results were published in four volumes titled **Wind Energy Resource Survey in India**. These volumes provided a detailed analysis of wind resources across the country, identifying promising sites for wind farms and providing the data needed to design wind energy projects.

India's Wind Energy Revolution

Today, India is a world leader in wind energy, with over 40 gigawatts of installed capacity—enough to power millions of homes. Wind farms dot the landscapes of Tamil Nadu, Gujarat, Maharashtra, and other states, their giant turbines turning gracefully in the wind.

This revolution was built on the foundations Anna Mani laid. Her wind surveys identified the best sites. Her data helped developers choose appropriate turbines. Her methods set standards that continue to be used.

Anna did not live to see the full flowering of Indian wind energy, she passed away in 2001, when installed capacity was still modest. But she would have been proud to know that her work helped make it possible.

The wind survey was Anna's last major project. In 1994, she suffered a stroke that left her partially immobilised. She continued to work as long as she could, but by 1996, her health forced her to retire from active research.

She spent her final years with her elder sister, Chellamma Chandy, in Thiruvananthapuram. Even in illness, she remained interested in science, in the world around her, in the progress of the projects she had started.

Anna Mani passed away on August 16, 2001, one week before her 83rd birthday. True to her convictions, she was cremated in the electric crematorium in Thiruvananthapuram- a final simplicity that matched the life she had lived.

CHAPTER 7

The Scientist and the Human Being

"We have only one life. First, equip yourself for the job, make full use of your talents and then love and enjoy the work, making the most of being out of doors and in contact with nature."

- Anna Mani

A Leader Who Cared

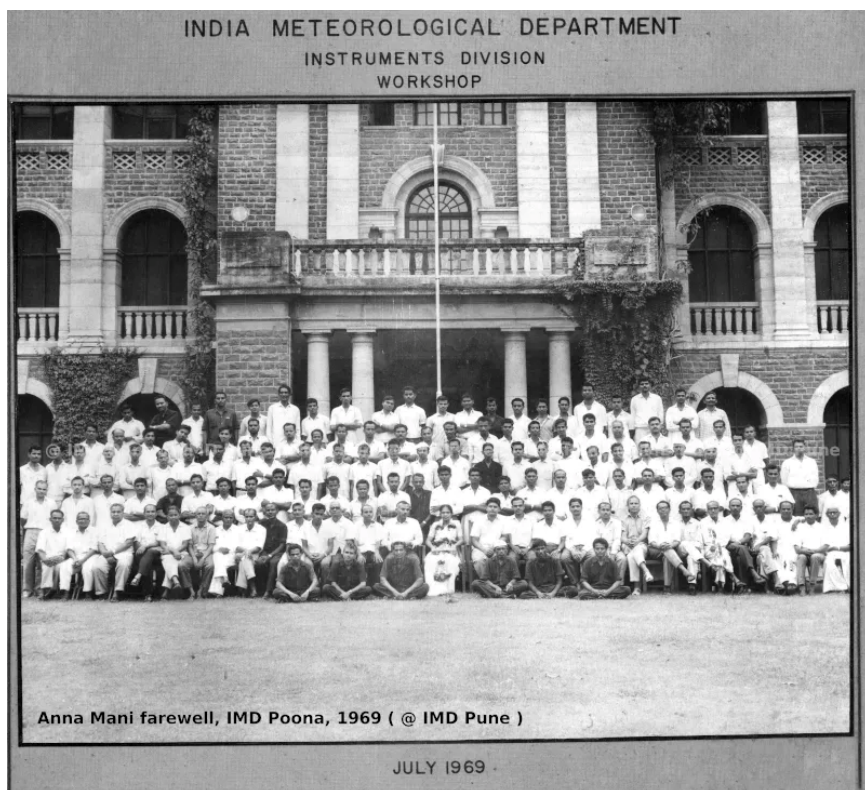
Throughout her career, Anna Mani was known as a demanding leader. She expected excellence from everyone who worked with her, and she accepted nothing less. Colleagues who made mistakes or missed deadlines quickly learned to fear her displeasure.

But there was another side to Anna, a side that those who knew her well came to appreciate. Beneath the demanding exterior was a deeply caring human being who looked after her colleagues with genuine concern.

G.P. Srivastava, a junior colleague who worked with Anna for many years, experienced this firsthand. When he was transferred to Delhi, Anna made sure he had warm clothing, knowing that someone from Pune might not be prepared for the colder Delhi winters. She checked on his welfare, made sure he was settled, and remained in touch throughout his career.

This was not an isolated incident. Anna's colleagues remember many such gestures: staying late to help someone struggling with a problem, providing resources for a promising project, and defending a junior scientist against unfair criticism.

Anna's leadership style was shaped by her own experiences. She had faced discrimination and dismissal as a young woman in Raman's laboratory. She had been denied a Ph.D. for reasons that had nothing to do with her



*(Figure 7.1: Anna Mani Farewell, IMD Poona, 1969,
Image Courtesy: <https://annamani.co.in/imd.html>)*

ability. She knew what it felt like to be undervalued, to have to prove yourself constantly.

As a leader, she was determined that others would not face the same obstacles. She gave her junior colleagues responsibility and trusted them to deliver. She celebrated their successes and supported them through failures. She created an environment where talent could flourish, regardless of background or circumstance.

The Letters to G.P. Srivastava

One of the most precious sources for understanding Anna Mani as a human being is her correspondence with G.P. Srivastava. These letters, spanning from the 1960s to the 1990s, were discovered in Anna's flat in Bangalore after her death by biographer Asha Gopinathan.

The letters reveal a woman who was not just a scientist but a fully rounded human being—observant, caring, witty, and deeply engaged with the world around her.

Anna wrote about her work, of course, about experiments, instruments, data, and publications. But she also wrote about her travels, describing the places she visited and the people she met. She wrote about Leningrad (now St. Petersburg), where she participated in international instrument comparisons, and about the warmth of her hosts. She wrote about Egypt, where she worked as a WMO consultant, and about colleagues who helped her find a place to stay.

Try to Imagine

You are Anna Mani, far from home, writing a letter to a junior colleague. Outside your window, the streets of Leningrad are covered with snow, a sight you never saw growing up in tropical Kerala. You have spent the day comparing radiation instruments with scientists from around the world. Now, in the quiet evening, you pick up your pen.

What do you write? Do you describe the snow, the people, the science? Do you offer advice to your younger colleague? Do you share your observations about this strange, cold country?

Anna's letters did all of this. They were windows into her world—and windows into her character.

The letters also revealed Anna's sense of humour. In one letter, she joked about a maid who cleaned her house so thoroughly that not even a bit of soap was left. Anna had to go out and buy everything again. This small, everyday incident became a delightful story, shared with a colleague who would appreciate the humour.

These letters matter because they show us Anna as a person, not just a scientist. They remind us that behind the published papers and professional achievements was a human being who laughed, who cared, who noticed the small things in life. They make her real to us in a way that biographical facts alone cannot.

Defender of a Writer

One of the most touching stories about Anna Mani involves her defence of a colleague who was also a writer. **C. Radhakrishnan** was a Malayalam writer

who worked at an IMD centre. His colleagues, apparently uncomfortable with his literary pursuits, arranged to have him transferred to Pune, hoping, perhaps, that the change would discourage his writing.

Instead, he encountered Anna Mani.

Anna read Radhakrishnan's books. She recognised their quality. She understood that writing was not a distraction from his work but an integral part of who he was. Instead of discouraging him, she supported him.

When his file crossed her desk, she wrote a note praising his literary achievements and recommending that he be supported, not penalised. Her intervention helped ensure that Radhakrishnan could continue both his scientific work and his writing. This story reveals something essential about Anna: she judged people by their quality, not by conformity. She understood that a person could have multiple passions, multiple talents, multiple contributions to make. She valued creativity in all its forms, not just in the narrow channels of scientific publication.

Mini-Profile: C. Radhakrishnan

C. Radhakrishnan is a renowned Malayalam writer, author of numerous novels, short stories, and travelogues. He worked for many years in the India Meteorological Department, balancing his scientific career with his literary passions.

His most famous work, *The Pandemic*, is a novel about the 1918 influenza outbreak—a subject that connects science and literature in a way that would have pleased Anna Mani.

Radhakrishnan never forgot Anna's support. In interviews, he has spoken of her with gratitude and admiration, describing her as a scientist who understood that science and art were not enemies but allies.

A Life Without Marriage

Anna Mani never married. In her community and era, this was unusual—almost all women married, usually in their late teens or early twenties. Anna's choice to remain single was a conscious one, reflecting her priorities and her philosophy.

When asked about it, she would laugh and say that taking care of herself was enough, one life was too short for all the knowledge she sought. This was

not bitterness or regret; it was a simple statement of fact. Marriage, for Anna, would have meant domestic responsibilities, compromises, and less time for science. She chose science.

But “chose” is perhaps too strong a word. Anna did not so much choose to remain single as she chose to live a life of purpose. That purpose was science. Everything else was secondary.

This does not mean Anna was isolated or lonely. She had friends and admirers around the world. She maintained close relationships with her siblings, particularly her elder sister Chellamma, who cared for her in her final years. She had colleagues who became lifelong friends, students who became like family. Her life was rich with human connection, even without marriage and children.

Anna’s attitude toward gender was characteristically straightforward. “My being a woman had absolutely no bearing on what I chose to do with my life,” she once said. “What is this hoopla around women and science?”

She refused to see herself as a victim or a pioneer. She was simply a scientist doing science. The fact that she was a woman did not matter, or at least, it should not have.

She did not want special treatment; she wanted equal opportunity, equal respect, equal recognition for equal work.

This attitude sometimes put her at odds with those who wanted to celebrate her as a woman scientist. She resisted such celebrations, not because she didn’t appreciate recognition, but because she believed that science should be judged on its merits, not on the gender of the scientist. Her work was good because it was good, not because a woman did it.

Trekking and Bird-Watching

Throughout her life, Anna maintained the love of nature she had developed as a child in Travancore. Whenever she had free time, she would escape to the outdoors—to trek, to watch birds, to simply be in nature.

This was not just recreation. It was a continuation of the observation skills she had honed as a child, and a reminder of why her work mattered. The atmosphere she studied was not an abstraction; it was the air she breathed, the sky she watched, the context for every living thing.

Bird-watching was a particular passion. Anna could identify dozens of species by sight and sound. She knew their habits, their habitats, their migrations. She understood that birds were indicators of environmental health, sentinels for changes that might otherwise go unnoticed.



(Figure 7.2: Anna Mani with her dogs Jenny and Sherry in Delhi, Image Courtesy: Anna Philipose, Image Source: <https://annamani.co.in/colleagues.html>)

Trekking took her to remote corners of India, to places she might otherwise never have seen. She climbed hills, crossed rivers, and walked through forests. Each trek was an adventure, an opportunity to see new things, to test her endurance, to connect with the land she loved.

Why Scientists Need Nature?

It might seem that a meteorologist, who studies the atmosphere, would be perfectly happy in a laboratory. But Anna Mani understood something important: science is not just about data; it's about understanding. And understanding comes from direct experience as well as measurement.

When you walk through a forest, you feel the humidity on your skin. When you climb a mountain, you experience the thinning air. When you watch a storm approach, you see the clouds building, feel the wind rising, smell the rain coming. These experiences inform your science, giving you an intuitive sense of how the atmosphere behaves.

Anna never lost this connection to direct experience. It kept her grounded, kept her curious, kept her human.

Anna also loved music and books. She was an avid reader throughout her life, continuing the habits of her childhood. Music, both classical and folk,

gave her joy and relaxation. These interests rounded out her life, providing balance to the intensity of scientific work.

The Final Years

In 1994, Anna suffered a stroke that left her partially paralysed. For someone who had been so active, so independent, this was a cruel blow. She could no longer trek, no longer travel, no longer work as she once had.

But she did not give up. With characteristic determination, she adapted. She continued to correspond with colleagues, to follow scientific developments, and to take an interest in the projects she had started. Her mind remained sharp even as her body failed.

Her sister Chellamma Chandy cared for her in Thiruvananthapuram, providing the support Anna needed. The two sisters, both elderly, shared a quiet life—reading, talking, remembering.

Anna Mani passed away on August 16, 2001, one week before her 83rd birthday. She was cremated in the electric crematorium in Thiruvananthapuram, a final simplicity that matched the life she had lived.

Chapter 8

Honours and Legacy

“To be a woman in science, you need to be strong and confident”.

- Anna Mani

Fellow of Academies

Throughout her career, Anna Mani received recognition from scientific academies around the world. These honours reflected the quality of her work and the respect she commanded among her peers.

In 1971, Anna was elected to the Council of the Indian Academy of Sciences, serving until 1979. She also served as one of the academy’s secretaries



(Figure 8.1: Anna Mani (Left) with Satish Dhawan (Second from Left), who guided the development of India’s space programme in its formative years, Image Source: Internet)

for three years and edited its newsletter, *Patrika*, from 1980 to 1995. This editorial work, often unnoticed, was crucial for communicating science to a wider audience.

In 1977, Anna was elected a Fellow of the Indian National Science Academy (INSA)—one of the highest honours for an Indian scientist. She served on the INSA Council from 1982 to 1984, contributing to the governance of Indian science.

Think Like a Scientist: What Does Fellowship Mean?

Scientific academies are organisations of distinguished scientists, dedicated to promoting science and recognising excellence. Fellowship is an honour conferred on scientists who have made significant contributions to their fields.

Fellows are elected by their peers, a rigorous process that ensures only the most deserving receive the honour. Being elected a Fellow means that the world's leading scientists consider your work worthy of recognition.

Anna's election to multiple academies- the Indian Academy of Sciences, Indian National Science Academy, and others, was a testament to the quality and impact of her work.

Anna was also a Fellow of several international organisations:

- American Meteorological Society
- American Geophysical Union
- Royal Meteorological Society (UK)
- Institution of Instrumentation and Telecommunication Engineers
- Solar Energy Society of India
- International Solar Energy Society

These fellowships recognized her contributions to meteorology, instrumentation, and renewable energy on the global stage. She was not just an Indian scientist; she was a world scientist.

The INSA K.R. Ramanathan Medal

In 1987, Anna received the INSA K.R. Ramanathan Medal, an award established in honour of her mentor, the man who had inspired her ozone work. The medal recognised her “outstanding contributions to atmospheric sciences and meteorology.”

INDIAN ACADEMY OF SCIENCES

BANGALORE 560 080

PARTICIPANTS TO THE 47TH ANNUAL MEETING
TRIVANDRUM, 5-8 NOVEMBER 1981



(Courtesy IAS)

(Figure 8.2: Participants to the 47th Annual Meeting of the Indian Academy of Sciences at Trivandrum, 5 – 8 November 1981.
Image Courtesy: Indian Academy of Sciences, Source: <https://annamani.co.in/conferences.html>)

Let Us Learn About: World Meteorological Organization, Indian Academy of Sciences & Indian National Science Academy

World Meteorological Organization (WMO)

The World Meteorological Organization (WMO) originated from the International Meteorological Organization (IMO), founded in 1873 to standardize weather observations. The WMO convention was signed on October 11, 1947, and officially ratified on March 23, 1950, becoming a UN specialized agency in 1951 to coordinate global weather, climate, and water data. The WMO maintains the principle of free, unrestricted data exchange, supporting its 193 member states and territories

Indian Academy of Sciences

The Indian Academy of Sciences Bangalore was founded by Sir C V Raman and registered as a society in 1934 with the aim of promoting the progress and upholding the cause of science, in both pure and applied branches. It strives to meet its objectives through promotion of original research and dissemination of scientific knowledge to the community via meetings, discussions, seminars, symposia and publications. Indian Academy of Sciences is now an autonomous institution under the Department of Science and Technology (DST) of the Government of India. It is fully funded by DST.

Indian National Science Academy (INSA)

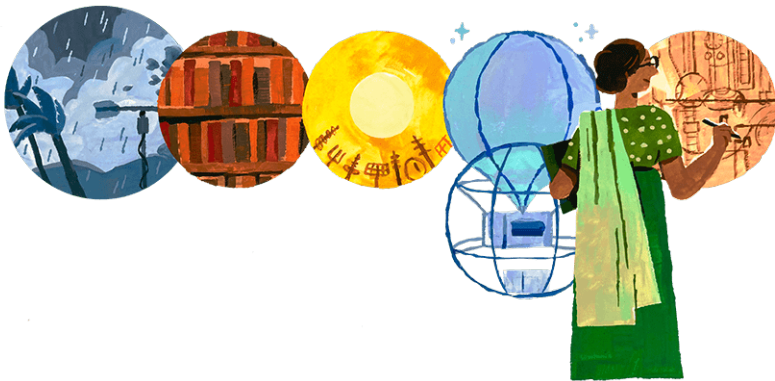
The National Institute of Sciences of India, now the Indian National Science Academy (INSA), was inaugurated on 7 January 1935 at Calcutta (now Kolkata) and functioned with its Headquarters at The Asiatic Society of Bengal till 1951 and thereafter shifted to Delhi. The Indian National Science Academy was established with the object of promoting science in India and harnessing scientific knowledge for the cause of humanity and national welfare. To act as a resource pool of knowledge; to identify and nurture scientific talent in the country, and to interface and represent the Indian Science and Scientists, globally on behalf of Govt. of India.

In these roles, she helped shape global standards for radiation measurement, organised international instrument comparisons, and coordinated activities across Asia. Her contributions were recognised by the WMO, which featured her in its bulletin and celebrated her on her 100th birth anniversary.

Anna also served as a WMO consultant in Egypt in 1975, training radiation staff at the Meteorological Office in Cairo. This work extended her influence beyond India, helping another developing country build its meteorological capability.

The International Ozone Commission welcomed Anna as a member, recognising her contributions to ozone measurement and her role in developing the Indian ozonesonde. She participated in international ozone symposia, contributed to global assessments, and helped build the scientific understanding that would lead to the Montreal Protocol.

The Google Doodle



(Figure 8.4: Google Doodle: Anna Mani's 104th Birthday, Image Source: <https://doodles.google/doodle/anna-manis-104th-birthday/>)

On August 23, 2022, what would have been Anna Mani's 104th birthday, Google honoured her with a doodle on its homepage. The colourful illustration showed Anna with weather instruments, surrounded by the sun, clouds, and wind that had been the focus of her work.

For one day, millions of people around the world saw Anna Mani's face. They read about her contributions, her struggles, her legacy. A scientist who had worked quietly for decades, who had never sought fame, was suddenly visible to everyone.

The Google doodle was a fitting tribute. It recognised not just Anna's scientific achievements but her role as an inspiration, especially for women and girls considering careers in science. It said, in effect: this woman mattered. Her work mattered. Her story deserves to be known.

Anna Mani's doodle on her 104th birthday brought her story to a new generation, ensuring that her legacy would continue to inspire.

Google Doodles

Google Doodles are special, temporary alterations of the Google logo on the search engine's homepage. They commemorate holidays, events, achievements, and notable people.

Since the first doodle in 1998 (commemorating the Burning Man festival), Google has created thousands of doodles for audiences worldwide. Doodles have honoured scientists, artists, activists, and leaders from every continent.

Being honoured with a Google doodle means that your story is considered important enough to share with millions of people. It's a form of recognition that reaches far beyond scientific circles—into schools, homes, and communities around the world.

Why Anna Mani Still Matters?

Anna Mani passed away in 2001, but her legacy lives on. In countless ways, her work continues to shape Indian science and Indian life.

The instruments she designed and standardised are still in use across India, measuring temperature, pressure, humidity, and wind. The calibration procedures she established ensure that measurements are accurate and comparable. The data she collected on solar radiation and wind energy informs renewable energy projects to this day.

The ozonesonde she developed continues to be flown from Indian stations, providing data on atmospheric ozone. These measurements contributed to the discovery of the ozone hole and continue to monitor its recovery. India remains one of the few countries conducting systematic ozone soundings from the tropics, a legacy of Anna's foresight.

Indian Academy of Sciences meeting in Ahmedabad, 1968 (Courtesy IAS)

RAMAN'S 80TH BIRTHDAY



(Figure 8.5: Anna Mani (sitting right to Sir C.V. Raman) with stalwarts of science and Technology in modern India during the 80th Birthday celebration of Sir C.V. Raman, 1968, Image Courtesy: Indian Academy of Sciences, Source: <https://annamani.co.in/conferences.html>)

Think Like a Historian: Why do we Remember?

We remember scientists for different reasons. Some we remember for a single great discovery—the Raman Effect, the theory of relativity, the structure of DNA. Others we remember for building institutions, for training students, for creating the infrastructure of science.

Anna Mani belongs to this second group. Her legacy is not a single discovery but a thousand small contributions that together transformed Indian meteorology. The instruments she standardised, the procedures she established, the people she trained, the networks she built—all of these continue to shape Indian science.

We remember Anna because she showed what dedication could achieve, and she overcame the obstacles that would have defeated any other person in normal circumstances. She served her nation – our India – our *Bhārata* with a distinction, with a pre eminence that her story inspires us to do the same.

The wind farms that dot the Indian landscape, their giant turbines turning gracefully in the breeze, are part of Anna's legacy. The solar panels on rooftops across India, converting sunlight into electricity, owe something to her work. The accurate weather forecasts that guide farmers, pilots, and fishermen depend on instruments she helped develop.

But perhaps most importantly, Anna's story continues to inspire. Young scientists, especially young women, look at her life and see what is possible. They see that you can come from a small town, face discrimination and setbacks, and still make a difference. They see that excellence, dedication, and integrity matter. They see that one person can change the world.

Anna Mani once said: “We have only one life. First equip yourself for the job, make full use of your talents, and then love and enjoy the work, making the most of being out of doors and in contact with nature.”

She lived those words. She equipped herself through years of study and hard work. She made full use of her talents, contributing to science and nation-building. She loved her work and found joy in it. And throughout her life, she remained connected to the natural world—the sky she studied, the wind she measured, the sun she harnessed.

That is her legacy. That is why Anna Mani still matters. And that is why her story will continue to inspire generations to come.

Closing Reflections: What Anna Mani Teaches Us

As we come to the end of this journey through Anna Mani's life, it is worth pausing to reflect on what her story teaches us. Anna was not a mythical

figure; she was a human being, with strengths and weaknesses, successes and setbacks. But her life offers lessons that are valuable for anyone, whatever path they choose.

1. Curiosity is a Lifelong Companion

Anna's curiosity never faded. From the eight-year-old who chose encyclopaedias over earrings to the seventy-year-old organising wind measurements across India, she remained endlessly interested in the world around her. She asked questions, sought answers, and never assumed she knew enough.

This curiosity was not limited to science. Anna read widely, listened to music, watched birds, and trekked through forests. She understood that knowledge is not a collection of separate facts but a connected whole, that understanding the atmosphere meant understanding the earth, the life it supported, and the people who lived within it.

Lesson for us: Cultivate curiosity. Ask questions. Never stop learning. The world is full of wonders waiting to be discovered.

2. Setbacks Are Not Endings

Anna faced numerous setbacks in her career. Denied a Ph.D. for lack of a master's degree, she did not abandon science but found another path. Struggling for two years to develop a working ozonesonde, she persisted through failure after failure. Facing discrimination, she focused on her work rather than on the injustice.

Each setback could have been an ending. Each could have discouraged her, turned her away from science, convinced her that the obstacles were too great. Instead, she treated setbacks as problems to be solved, not as reasons to give up.

Lesson for us: Failure is part of the journey. What matters is not whether you fall, but whether you get up and keep going.

3. Serve Something Larger Than Yourself

Anna's work was driven by more than personal ambition. She wanted to serve India, to make the nation self-reliant, to build institutions that would last, to contribute to the well-being of her fellow citizens. This sense of purpose gave her work meaning and sustained her through difficult times.

This does not mean that Anna was indifferent to recognition or advancement. She valued both. But they were not her primary motivations. What drove her was the conviction that her work mattered - that accurate

weather measurements would help farmers, that solar data would enable renewable energy, that ozone monitoring would protect life on Earth.

Lesson for us: Find a purpose larger than yourself. Work for something that will outlast you. Contribute to the common good.

4. Lead by Example

As a leader of 121 men, Anna could have relied on authority alone. Instead, she led by example - working as hard as anyone, mastering every detail, holding herself to the highest standards. She earned respect through competence and dedication, not through position or power.

Anna's leadership style was not about being liked; it was about being effective. She was demanding because excellence mattered. She was fair because justice mattered. She was caring because people mattered. Her team knew that she would support them, but also that she would expect their best.

Lesson for us: If you want others to give their best, give your best first. Lead not from above, but from within. Earn respect through action, not words.

5. Stay Humble

Despite all her achievements- the instruments standardised, the papers published, the honours received- Anna remained humble. She did not seek fame or recognition. She did not boast about her accomplishments. She simply did her work, day after day, year after year.

This humility was not false modesty. Anna knew the value of her work, but she also knew that science is a collective endeavour, built on the contributions of many, meant to serve the common good. She stood on the shoulders of those who came before, and she hoped that those who came after would stand on hers.

Lesson for us: Accomplishment does not require arrogance. The best work is often done quietly, by people who care more about the work than about the credit.

6. Integrate Science and Life

Anna never separated her science from her life. Her love of nature inspired her meteorology. Her reading enriched her thinking. Her relationships sustained her through challenges. Science was not a job she did from nine to five; it was a way of being in the world.

This integration kept Anna balanced. She had interests beyond science, relationships beyond work, pleasures beyond achievement. When her health failed, she had other resources to draw on—memories of treks, love of books, the companionship of family.

Lesson for us: Don't let your work consume you. Cultivate interests beyond your profession. Nurture relationships. Find joy in nature, in art, in simple pleasures. A full life supports good work.

7. Be Yourself

Perhaps the most important lesson from Anna's life is simply this: be yourself. Anna did not try to fit anyone's mould. She did not become what others expected. She followed her curiosity, pursued her passions, and lived according to her values.

This authenticity was not always easy. Anna faced pressure to conform—to marry, to dress conventionally, to accept the roles assigned to women. She refused. She wore khadi because it expressed her values. She remained single because marriage would have constrained her work. She led men because she was competent, regardless of what anyone thought.

Lesson for us: The path of authenticity is not always smooth, but it is the only path to a life fully lived. Discover who you are. Honour that. Let your life express your deepest values.

A Final Word

Anna Mani's life was extraordinary, but not in the way we usually think of extraordinary. She did not discover a new law of nature or win a Nobel Prize. She did something perhaps more difficult: she built the infrastructure of science in a newly independent nation. She created the instruments, established the standards, trained the people, and laid the foundations on which others could build.

Her legacy is all around us—in every weather forecast, every solar panel, every wind farm. But it is also within us, in the inspiration her story provides. When we face obstacles, we can remember Anna. When we doubt ourselves, we can remember Anna. When we wonder whether one person can make a difference, we can remember Anna.

She showed us what is possible. Now it is up to us to carry her legacy forward.

APPENDICES

Appendix – A: Timeline of Anna Mani's Life

Year	Event
1918	Born August 23 in Peermade, Travancore (now Kerala)
1926	Chooses Encyclopaedia Britannica over diamond earrings
1930	Reads every book in the local library
1935	Enrolls at Pachaiyappas College, Madras
1937	Transfers to Presidency College, Madras
1939	Graduated with B.Sc. (Honours) in Physics and Chemistry
1940	Awarded a research scholarship to IISc, Bangalore
1940-45	Works in C.V. Raman's laboratory
1942	Published first paper on ruby spectroscopy
1944	Published three papers on diamond spectroscopy
1945	Published fifth paper; submits Ph.D. dissertation
1945	Ph.D. dissertation rejected by the University of Madras
1945	Awarded a government scholarship for England
1945	Sails to England on a troop ship
1945-48	Studies meteorological instrumentation in the UK
1948	Returns to India, joins the India Meteorological Department
1948-53	Leads the development of indigenous instruments
1953	Takes charge of the Instruments Division (121 staff)
1953-57	Establishes primary standards, standardises instruments
1957	Begins solar radiation work during the IGY
1957-60	Leads metric system conversion
1957-69	Builds a national radiation network
1960	Begins studying atmospheric ozone
1962	Takes on the challenge of developing the Indian ozonesonde

Year	Event
1963	Set up a meteorological observatory and an instrumentation tower at the Thumba rocket launching facility, at the request of Vikram Sarabhai
1964	First successful ozone sounding from Pune
1965-70	Refines ozonesonde, expands network
1969	Pune recognised as the Regional Radiation Centre
1970	Indian ozonesonde succeeds in international comparison
1971	Elected to the Council, Indian Academy of Sciences
1976	Retires from IMD
1977	Elected Fellow, Indian National Science Academy
1980	Published Handbook of Solar Radiation Data
1981	Published Solar Radiation over India
1982	Indian Antarctic expeditions begin ozone soundings
1982-84	Serves on INSA Council
1984	Begins wind energy survey
1985	Farman published the discovery of ozone hole
1987	Received INSA K.R. Ramanathan Medal
1990-94	Published four volumes on wind energy
1994	Suffers stroke
1996	Retires from wind survey
2001	Passes away on August 16 in Thiruvananthapuram
2022	Google Doodle on 104th birth anniversary

Appendix – B:

Major Publications

Research Papers (Selected)

1. Fluorescent absorption and scattering of light in Ruby. (1942). Anna Mani. *Proceedings of the Indian Academy of Sciences*. Sci XV P52
2. (With Venkiteswaran S.P. and Amritan K.G.) (1954) A tipping bucket type of distant recording intensity raingauge *Indian Journal of Meteorology and Geophysics*. 5 No. 3 61-64
3. (With Venkiteswaran, S.P.). Radar studies of rain at Poona. (1961). *Proc Weather Radar Conference*, 9th Kansas City, 404-409
4. (With Sreedharan, C.R.) “Measurement of the vertical distribution of ozone by a chemical ozonesonde” (1965). *Current Science*. 34 No. 2. 39-4.
5. (With Sreedharan, C.R.) “Variations in the vertical distribution of ozone over India” (1969). *Annales de Geophysique*. 25. No. 1 173- 181.
6. (With Chacko, O.) “Solar radiation climate of India” (1973). *Solar Energy*. 14. 139 – 156.
7. (With Chacko, O. and Iyer, N.V.) “Atmospheric turbidity over India from solar radiation measurements” (1973). *Solar Energy*. 14. 185 – 195.
8. (With Rangarajan, S.) “Variations in the observed total ozone distributions in the tropics” (1976). *Proceedings of the Joint Symposium on Atmospheric Ozone*. Dresden 1. 229-247.
9. (With Sikka D.R. et al.) Radiometersounding over the Indian Region during the summer Monex 1979. (1980). *Intl Symposium on Radiation*. Fort Collins.

Books

1. **Handbook of Solar Radiation Data for India** (1980). New Delhi: Allied Publishers.
2. **Solar Radiation over India** (1981, with S. Rangarajan). New Delhi: Allied Publishers.
3. **Wind Energy Data for India** (1983, with D.A. Modey). New Delhi: Allied Publishers.
4. **Wind Energy Resource Survey in India** (Vol. I) (1990). New Delhi: Allied Publishers.
5. **Wind Energy Resource Survey in India** (Vol. II) (1992). New Delhi: Allied Publishers.
6. **Wind Energy Resource Survey in India** (Vol. III & IV) (1994 with S. Rangarajan). New Delhi: Allied Publishers.

Appendix – C

Glossary of Scientific Terms

Term	Meaning
Absorption spectrum	Pattern of wavelengths absorbed by a material, revealing its composition and structure
Anemometer	Instrument for measuring wind speed
Barograph	A recording barometer that produces a continuous pressure record
Barometer	Instrument for measuring atmospheric pressure
Bose-Einstein statistics	Quantum statistics describing particles that can occupy the same state (bosons)
Calibration	Process of comparing an instrument's readings to known standards
Chlorofluorocarbons (CFCs)	Chemicals that deplete the ozone layer, used in refrigeration and aerosols
Crystallography	Study of crystal structure and properties
Dobson spectrophotometer	Instrument for measuring total atmospheric ozone
Fluorescence	Emission of light by a substance after absorbing radiation
Hygrograph	Recording hygrometer for continuous humidity record
Hygrometer	Instrument for measuring humidity
International Geophysical Year (IGY)	Global scientific effort (1957-58) to study the Earth and atmosphere
Meteorology	Scientific study of the atmosphere and weather
Montreal Protocol	International treaty phasing out ozone-depleting substances

Appendix – D:

Further Reading and Exploration

Those seeking a deeper understanding of the life and enduring contributions of Anna Mani may consult the following carefully curated selection of websites, digital archives, and scholarly works. Collectively, these resources illuminate her intellectual trajectory, collaborative ethos, and sustained impact on scientific research, offering meaningful insights for both scholars and general readers alike.

- Sreedharan, C. R. (2004). “Anna Modayil Mani”. Biog. Mem. Fell. INSA, N. Delhi, 25, 77-91. (http://www.insaindia.res.in/BM/BM25_0408.pdf)
- Sur, A. “An appreciation of Anna Mani”. Indian Academy of Sciences. (https://www.ias.ac.in/public/Resources/Initiatives/Women_in_Science/Contributors/annamani.pdf).
- Gopinathan, A. (2025). *Anna Mani: The Uncut Diamond*. National Book Trust, India, 2025 - (ISBN:93-6719-480-3, ISBN 13:978-93-6719-480-5)
- Taba, H. “The Bulletin Interviews”. WMO-No 865, pp. 139-149. <https://www.scribd.com/document/542833151/Miss-Anna-Mani-Bulletin-Article-1991>. (https://library.w-mo.int/pmb_ged/wmo_865_en.pdf)
- Vigyan Prasar. “Weathering Heights: Anna Mani”. (<https://www.youtube.com/watch?v=b1F517dt71k&t=25s>)
- Gupta, A. (2012). “Anna Mani”. Platinum Jubilee Publishing of INSA. (<http://www.arvindguptatoys.com/arvindgupta/bs30annamani.pdf>)
- Anna Mani: The Trailblazing Physicist <https://annamani.co.in/index.html>

ABOUT THE BOOK

The book, *Anna Mani: The Weather Woman of India*, presents a thoughtful and engaging account of the life and scientific legacy of Anna Mani, one of India's pioneering meteorologists. It traces her intellectual journey from a curious child deeply connected with nature to a scientist who played a crucial role in developing indigenous meteorological instrumentation and atmospheric research in India. The narrative highlights her commitment to scientific self-reliance, precision in observation, and quiet yet transformative leadership. Carefully designed for young readers as well as scholars, it combines scientific insight with human depth, encouraging readers to appreciate inquiry, resilience, and the ethical dimensions of science as a meaningful pursuit.

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